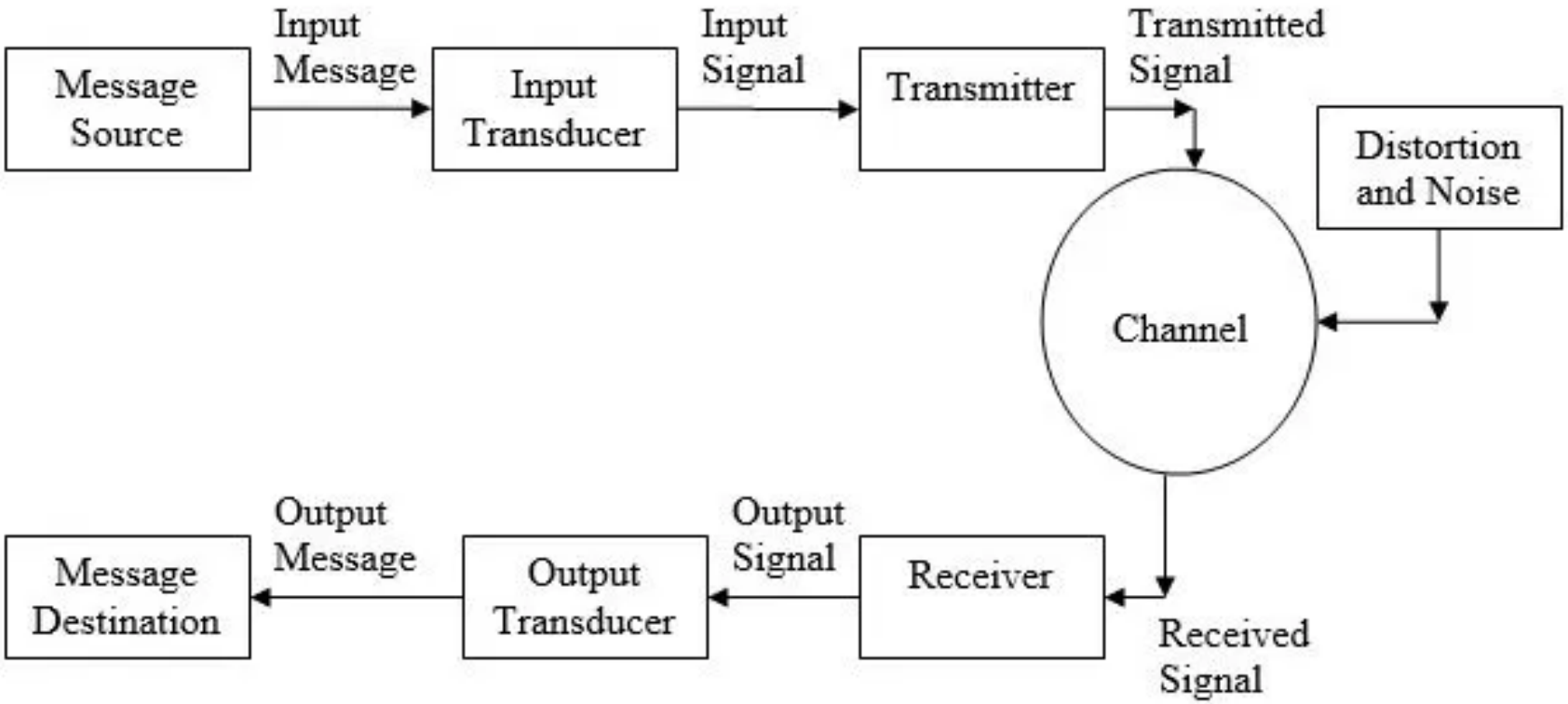
2

THE PHYSICAL LAYER

Fundamental Model of Communication Systems



Fundamental Model of Communication Systems

- **Message Source**: The originator of the message
- **Input Message**: The message/data/info that is to be communicated
- **Input Transducer**: Converts the input message into electrical form
- **Input Signal**: The data in electrical form (this is a baseband signal)
- **Transmitter**: Modifies the signal for transmission
- **Channel**: The medium over which the transmitted signal is sent (e.g., wire, air, optical fiber, free space)
- **Distortion/Noise**: External signals/features that affect the signal
- **Receiver**: Modifies the received signal, undoing the modifications done by the transmitter
- **Output Transducer**: Converts message from electrical signal back into its original form
- **Output Message**: The message/data/info that has been communicated
- **Message Destination**: Who/what the message/data/info was intended for

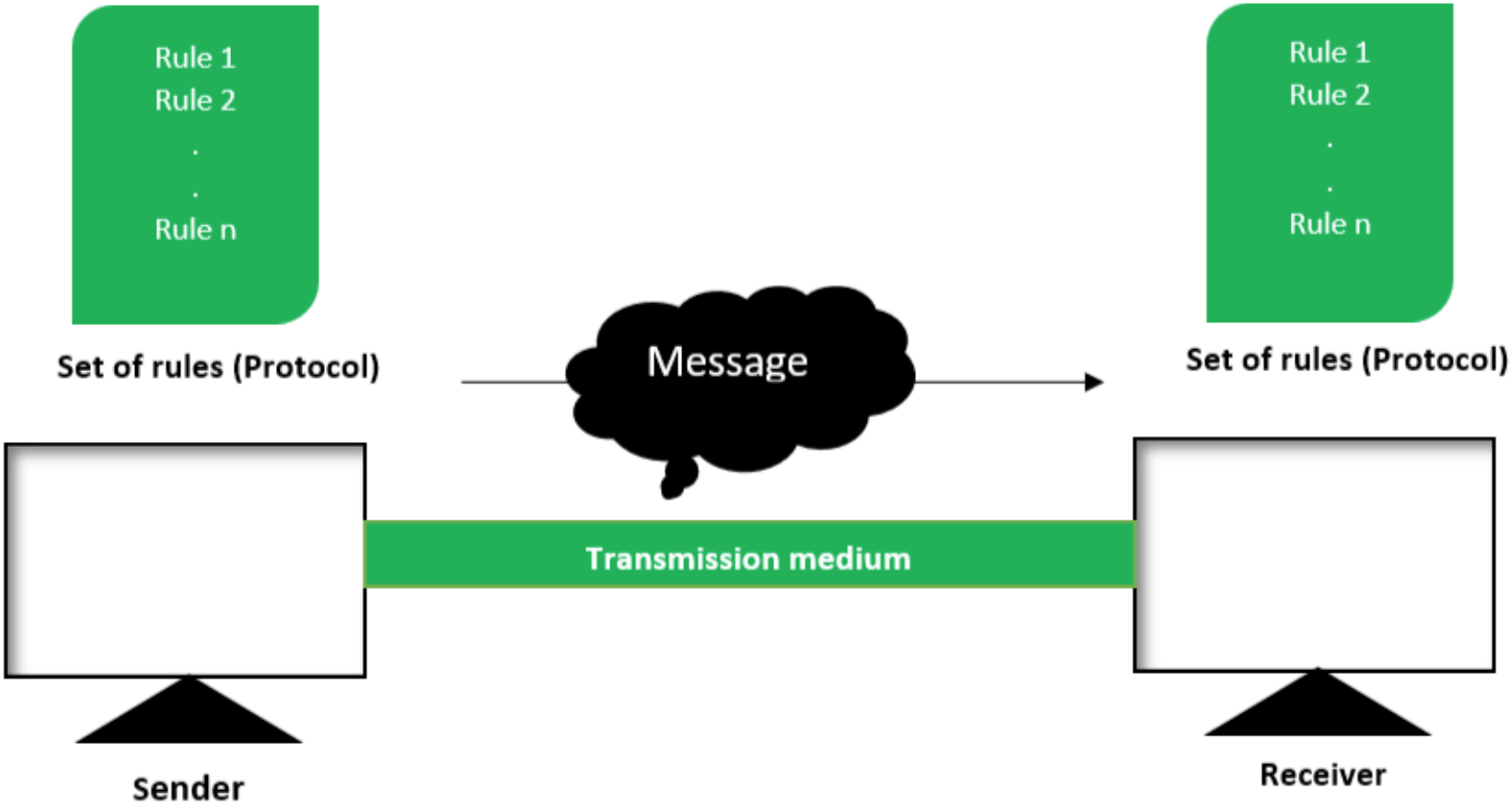
Introduction to Communication System

Components of Data Communication System

- Data Communication is defined as exchange of data between two devices via some form of transmission media such as a cable, wire or it can be air or vacuum also.
- For occurrence of data communication, communicating devices must be a part of communication system made up of a combination of hardware or software devices and programs.
- **Data Communication System Components :**
  There are mainly five components of a data communication system:

1. Message **2.** Sender **3.** Receiver **4.** Transmission Medium **5.** Set of rules (Protocol)

Introduction to Communication System



Introduction to Communication System

- **Message :**
  This is most useful asset of a data communication system. The message simply refers to data or piece of information which is to be communicated. A message could be in any form, it may be in form of a text file, an audio file, a video file, etc.
- **Sender :**
  To transfer message from source to destination, someone must be there who will play role of a source. Sender plays part of a source in data communication system. It is simple a device that sends data message. The device could be in form of a computer, mobile, telephone, laptop, video camera, or a workstation, etc.
- **Receiver :**
  It is destination where finally message sent by source has arrived. It is a device that receives message. Same as sender, receiver can also be in form of a computer, telephone mobile, workstation, etc.

Introduction to Communication System

- **Transmission Medium :**
  In entire process of data communication, there must be something which could act as a bridge between sender and receiver, Transmission medium plays that part. It is physical path by which data or message travels from sender to receiver. Transmission medium could be guided (with wires) or unguided (without wires), for example, twisted pair cable, fiber optic cable, radio waves, microwaves, etc.
- **Set of rules (Protocol) :**
  To govern data communications, various sets of rules had been already designed by the designers of the communication systems, which represent a kind of agreement between communicating devices. These are defined as protocol. In simple terms, the protocol is a set of rules that govern data communication. If two different devices are connected but there is no protocol among them, there would not be any kind of communication between those two devices. Thus the protocol is necessary for data communication to take place.
- Ex. – e-mail

Introduction to Communication System

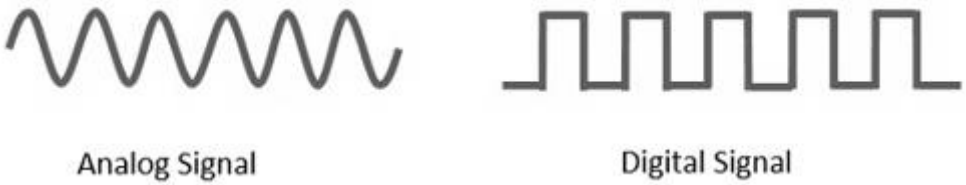
Digital Communication - Analog to Digital

- The communication that occurs in our day-to-day life is in the form of signals. These signals, such as sound signals, generally, are analog in nature. When the communication needs to be established over a distance, then the analog signals are sent through wire, using different techniques for effective transmission.

The Necessity of Digitization

- The conventional methods of communication used analog signals for long distance communications, which suffer from many losses such as distortion, interference, and other losses including security breach.
- In order to overcome these problems, the signals are digitized using different techniques. The digitized signals allow the communication to be more clear and accurate without losses.
- The following figure indicates the difference between analog and digital signals. The digital signals consist of **1s** and **0s** which indicate High and Low values respectively.

Introduction to Communication System



Representation of Signals

Introduction to Communication System

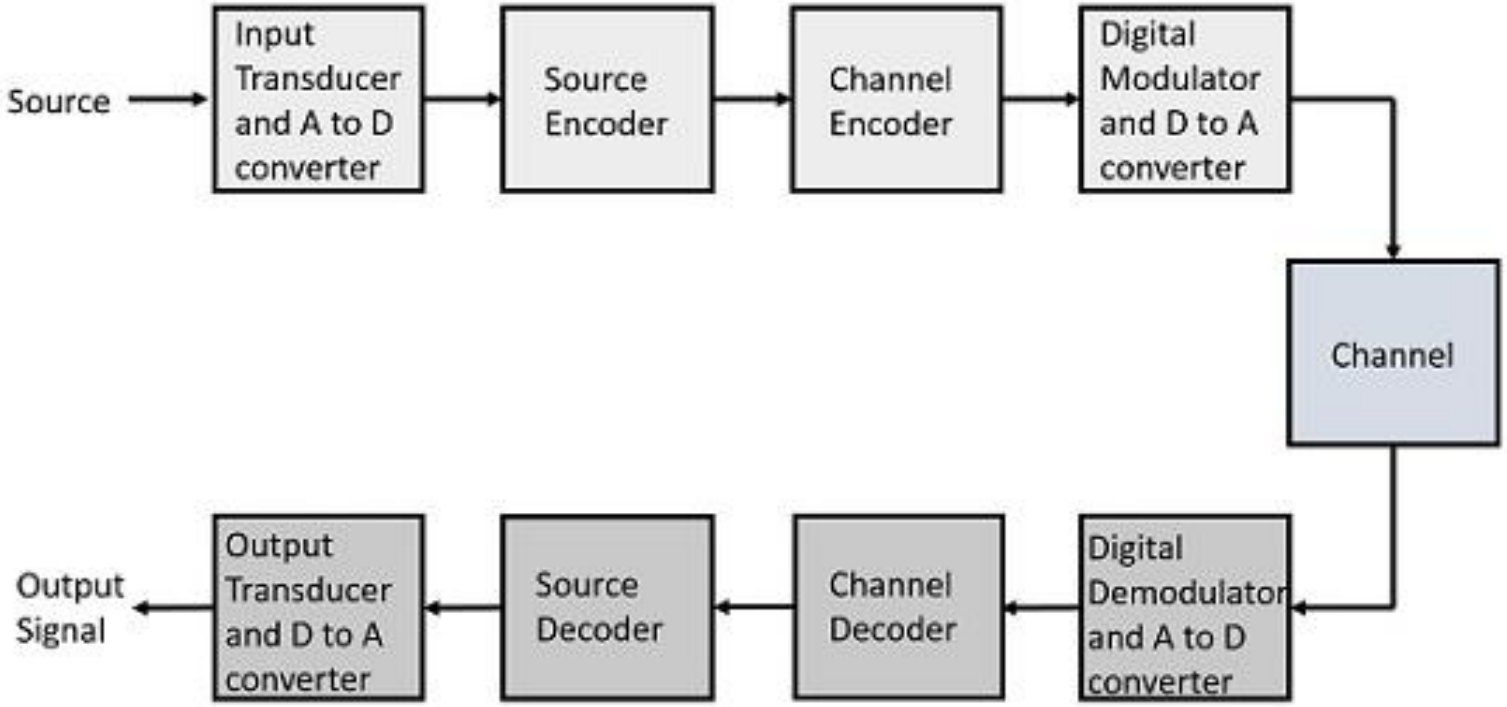
Advantages of Digital Communication

- As the signals are digitized, there are many advantages of digital communication over analog communication, such as −
- The effect of distortion, noise, and interference is much less in digital signals as they are less affected.
- Digital circuits are more reliable.
- Digital circuits are easy to design and cheaper than analog circuits.
- The hardware implementation in digital circuits, is more flexible than analog.
- The occurrence of cross-talk is very rare in digital communication.
- The signal is un-altered as the pulse needs a high disturbance to alter its properties, which is very difficult.
- Signal processing functions such as encryption and compression are employed in digital circuits to maintain the secrecy of the information.
- The probability of error occurrence is reduced by employing error detecting and error correcting codes.

Introduction to Communication System

- Spread spectrum technique is used to avoid signal jamming.
- Combining digital signals using Time Division Multiplexing TDMTDM is easier than combining analog signals using Frequency Division Multiplexing FDMFDM.
- The configuring process of digital signals is easier than analog signals.
- Digital signals can be saved and retrieved more conveniently than analog signals.
- Many of the digital circuits have almost common encoding techniques and hence similar devices can be used for a number of purposes.
- The capacity of the channel is effectively utilized by digital signals.

Elements of Digital Communication



Basic Elements of a Digital Communication System

Elements of Digital Communication

- **Source**

  The source can be an **analog** signal. **Example**: A Sound signal

- **Input Transducer**

  This is a transducer which takes a physical input and converts it to an electrical signal (**Example**: microphone). This block also consists of an **analog to digital** converter where a digital signal is needed for further processes.

  A digital signal is generally represented by a binary sequence.

- **Source Encoder**

  The source encoder compresses the data into minimum number of bits. This process helps in effective utilization of the bandwidth. It removes the redundant bits unnecessary excess bits, i.e. zeroes

- **Channel Encoder**

  The channel encoder, does the coding for error correction. During the transmission of the signal, due to the noise in the channel, the signal may get altered and hence to avoid this, the channel encoder adds some redundant bits to the transmitted data. These are the error correcting bits.

Elements of Digital Communication

- **Digital Modulator**

  The signal to be transmitted is modulated here by a carrier. The signal is also converted to analog from the digital sequence, in order to make it travel through the channel or medium.

- **Channel**

  The channel or a medium, allows the analog signal to transmit from the transmitter end to the receiver end.

- **Digital Demodulator**

  This is the first step at the receiver end. The received signal is demodulated as well as converted again from analog to digital. The signal gets reconstructed here.

- **Channel Decoder**

  The channel decoder, after detecting the sequence, does some error corrections. The distortions which might occur during the transmission, are corrected by adding some redundant bits. This addition of bits helps in the complete recovery of the original signal.

Elements of Digital Communication

- **Source Decoder**

  The resultant signal is once again digitized by sampling and quantizing so that the pure digital output is obtained without the loss of information. The source decoder recreates the source output.

- **Output Transducer**

  This is the last block which converts the signal into the original physical form, which was at the input of the transmitter. It converts the electrical signal into physical output (**Example**: loud speaker).

- **Output Signal**

  This is the output which is produced after the whole process. **Example** – The sound signal received.

Analog and Digital Data

- ***Data***

- ***Analog:*** refers to information that is continuous e.g., analog clock

- ***Digital:*** refers to information that has discrete states e.g., digital clock

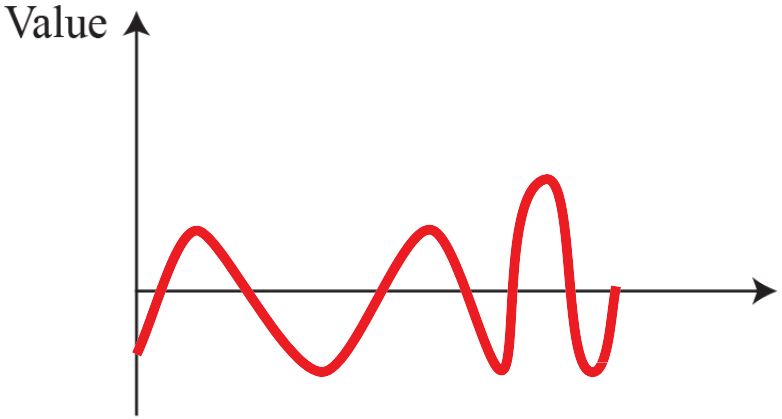
Analog and Digital Signals

Like the data they represent, signals can be either analog or digital.

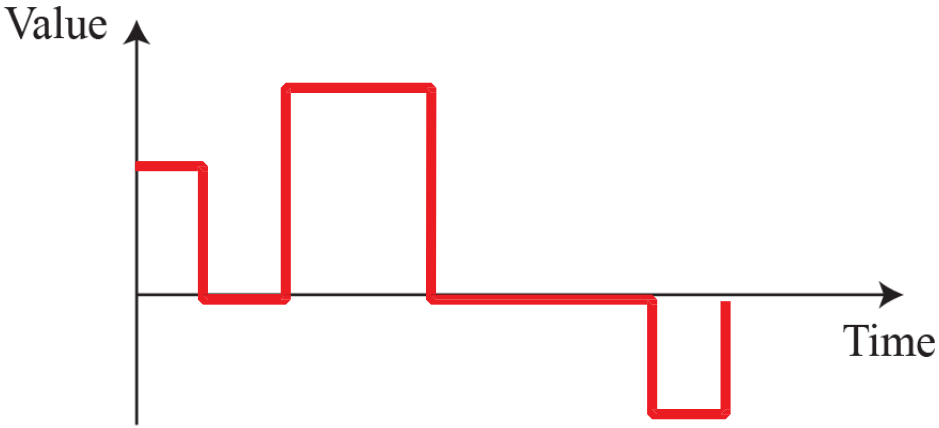
- *An analog signal has infinitely many levels of intensity over a period of time*

- *A digital signal can have only a limited number of defined values.*

Comparison of analog and digital signals



a. Analog signal

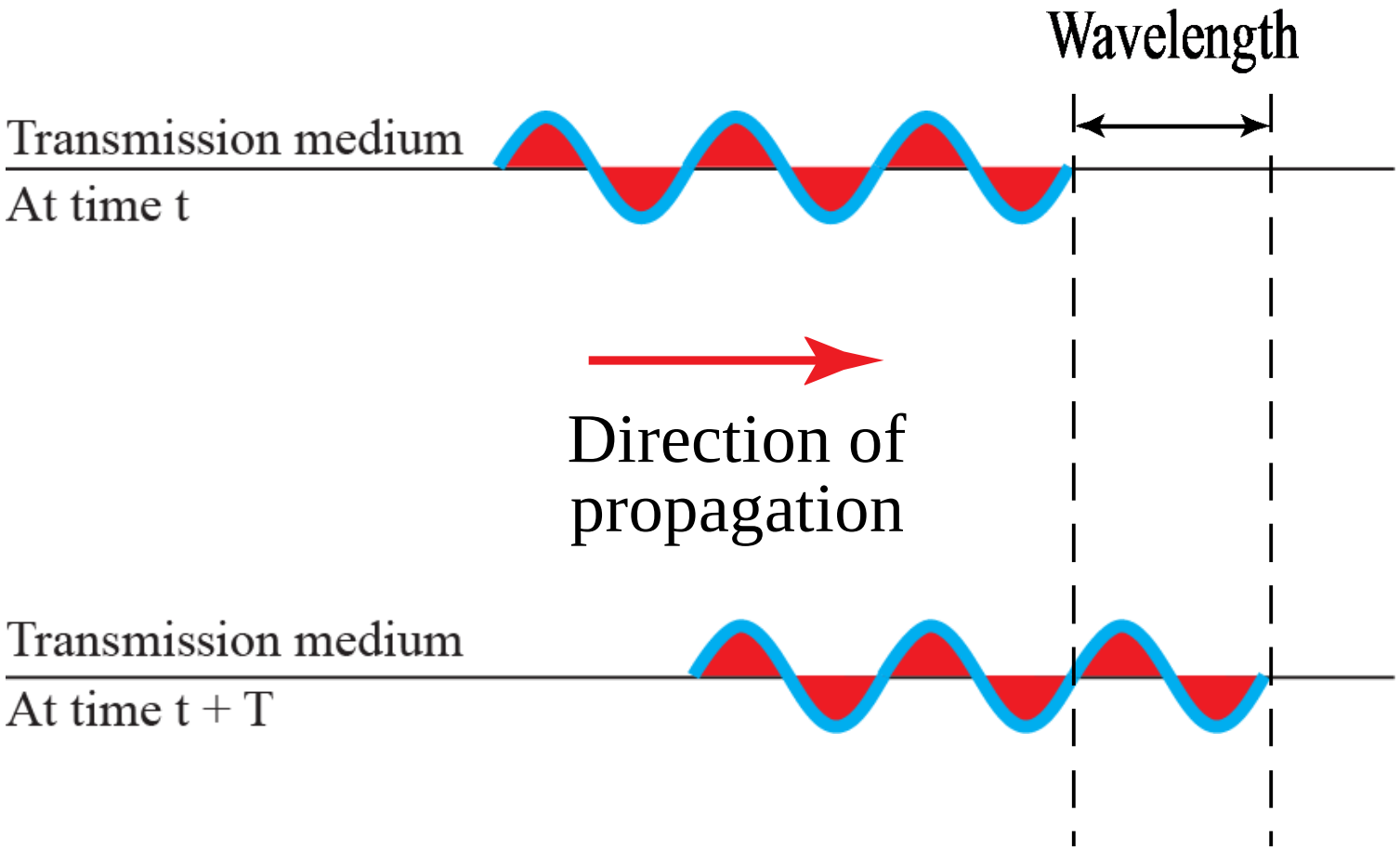


b. Digital signal

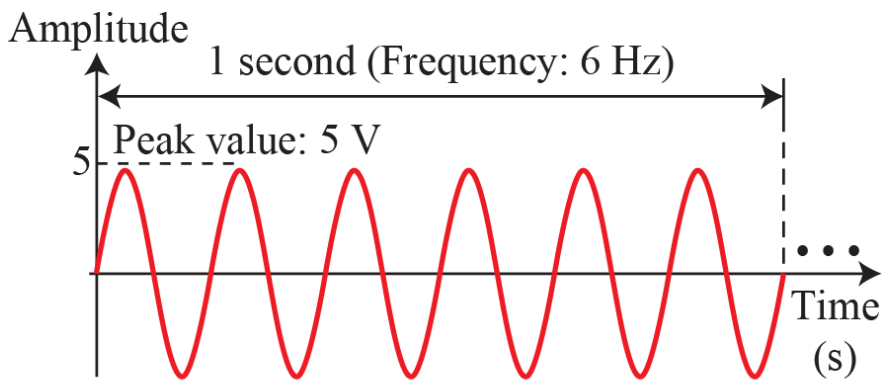
Units of period and frequency

Period		*Frequency*	
Unit	*Equivalent*	*Unit*	*Equivalent*
Seconds (s)	1 s	Hertz (Hz)	1 Hz
Milliseconds (ms)	10^{-3} s	Kilohertz (kHz)	10^{3} Hz
Microseconds (μs)	10^{-6} s	Megahertz (MHz)	10^{6} Hz
Nanoseconds (ns)	10^{-9} s	Gigahertz (GHz)	10^{9} Hz
Picoseconds (ps)	10^{-12} s	Terahertz (THz)	10^{12} Hz

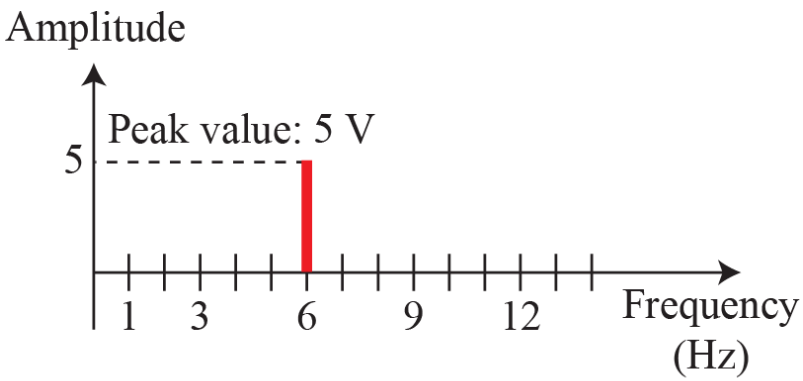
Wavelength and period



The time-domain and frequency-domain plots of a sine wave



a. A sine wave in the time domain

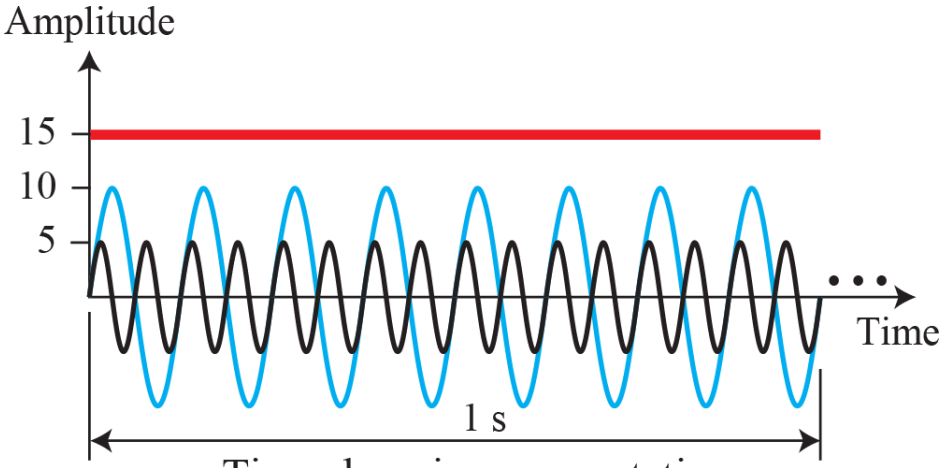


b. The same sine wave in the frequency domain

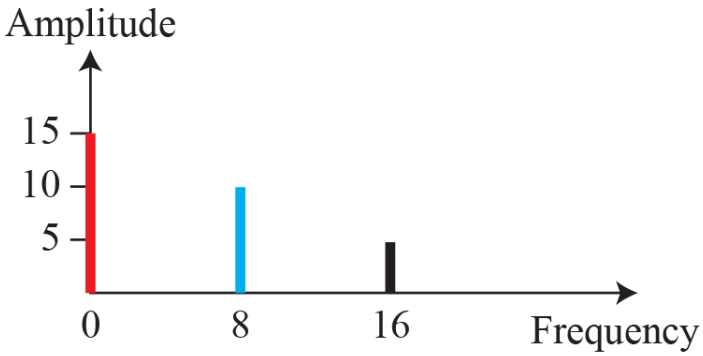
Example

The frequency domain is more compact and useful when we are dealing with more than one sine wave. For example, Figure 3.9 shows three sine waves, each with different amplitude and frequency. All can be represented by three spikes in the frequency domain.

The time domain and frequency domain of three sine waves



a. Time-domain representation



b. Frequency-domain representation

Bandwidth

The bandwidth is normally a difference between two numbers. For example, if a composite signal contains frequencies between 1000 and 5000, its bandwidth is 5000 − 1000, or 4000.

The width of the frequency range transmitted without being strongly attenuated is called the **bandwidth.**
In practice, the cutoff is not really sharp, so often the quoted bandwidth is from 0 to the frequency at which the received power has fallen by half.

Example

If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900 Hz, what is its bandwidth?

Solution

Let f_h be the highest frequency, f_l the lowest frequency, and B the bandwidth. Then

$$B = f_h - f_l = 900 - 100 = 800 \text{ Hz}$$

Example

A periodic signal has a bandwidth of 20 Hz. The highest frequency is 60 Hz. What is the lowest frequency? Draw the spectrum if the signal contains all frequencies of the same amplitude.

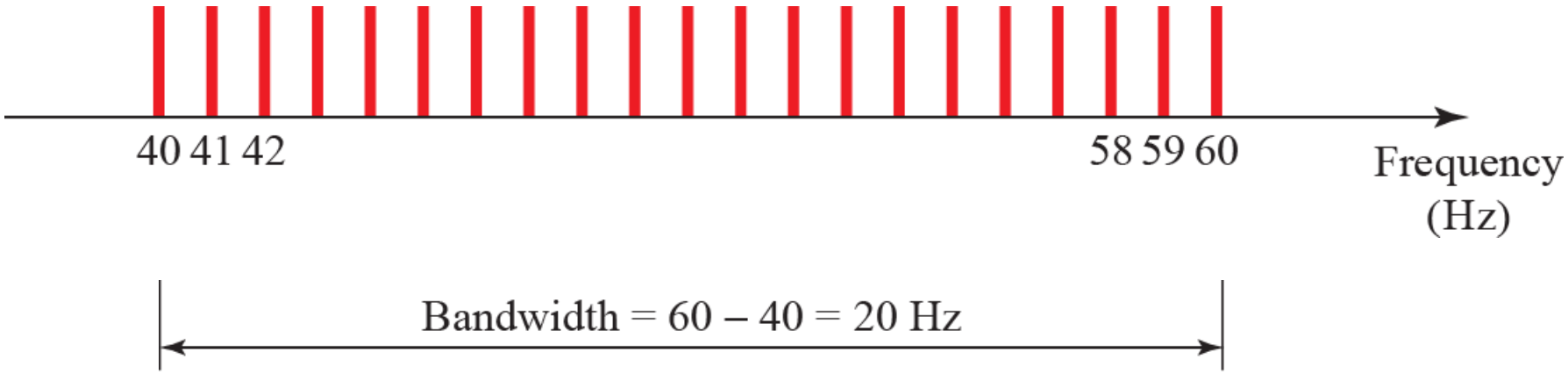
Solution

Let f_h be the highest frequency, f_l the lowest frequency, and B the bandwidth. Then

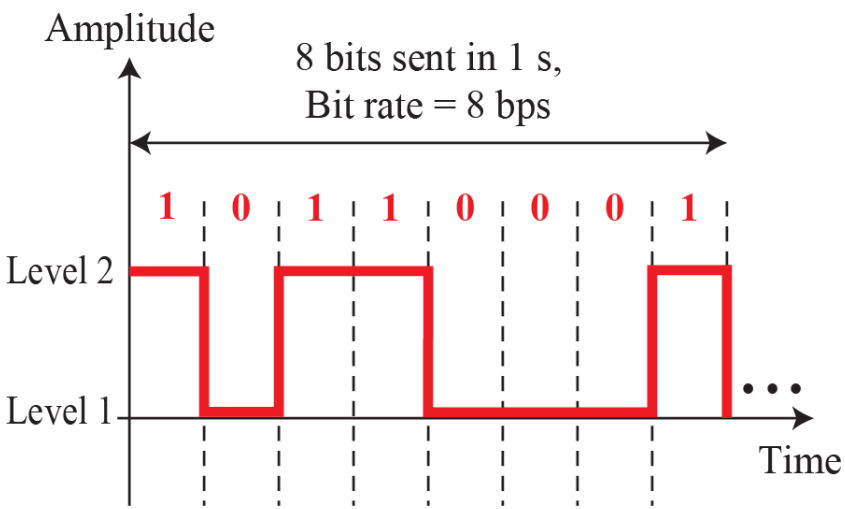
$$B = f_h - f_l \longrightarrow 20 = 60 - f_l \longrightarrow f_l = 60 - 20 = 40 \text{ Hz}$$

The spectrum contains all integer frequencies. We show this by a series of spikes

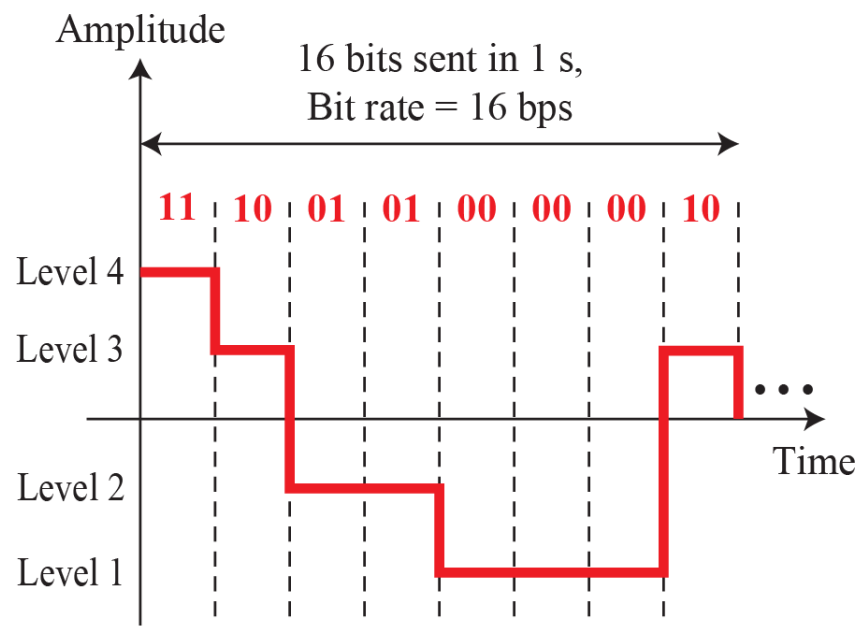
The bandwidth



Two digital signals: one with two signal levels and the *other with four signal levels*



a. A digital signal with two levels



b. A digital signal with four levels

Transmission Impairments

- *What is sent is not what is received*

- *Three causes of impairment are attenuation, distortion, and noise*

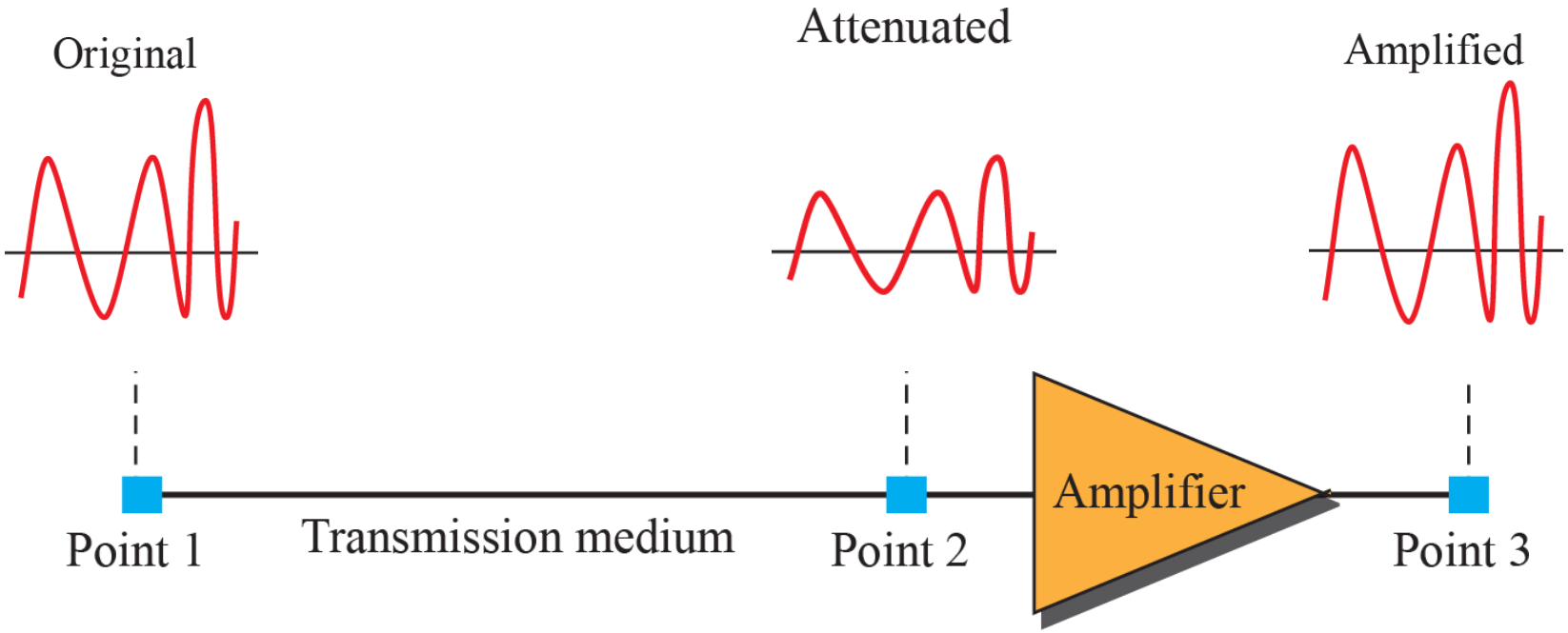
Attenuation

- *Attenuation means a loss of energy.*

- *When a signal, simple or composite, travels through a medium, it loses some of its energy in overcoming the resistance of the medium.*

- *To compensate for this loss, amplifiers are used to amplify the signal.*

Attenuation and amplification



Distortion

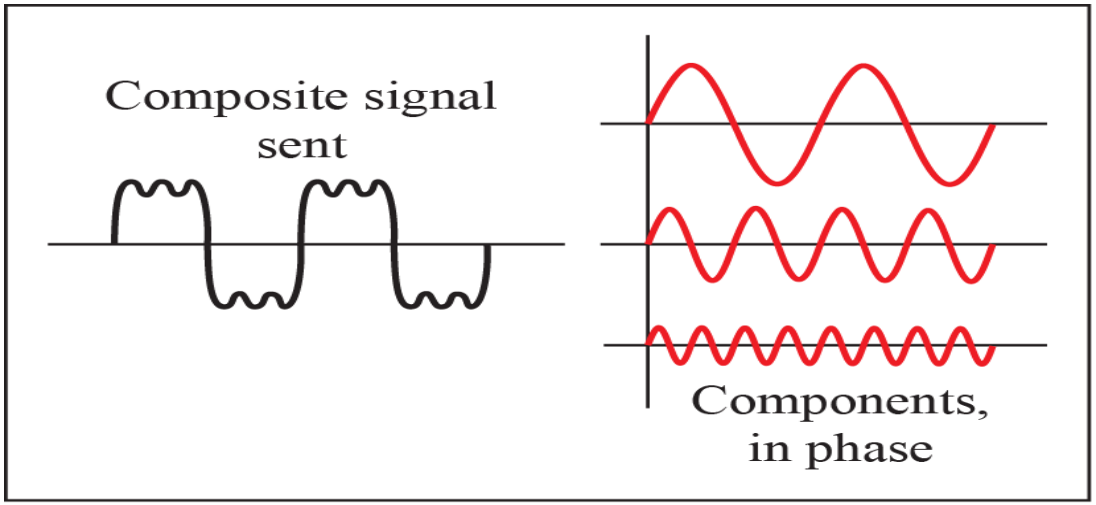
- *Signal changes its form or shape.*

- *Distortion can occur in a composite signal made of different frequencies.*

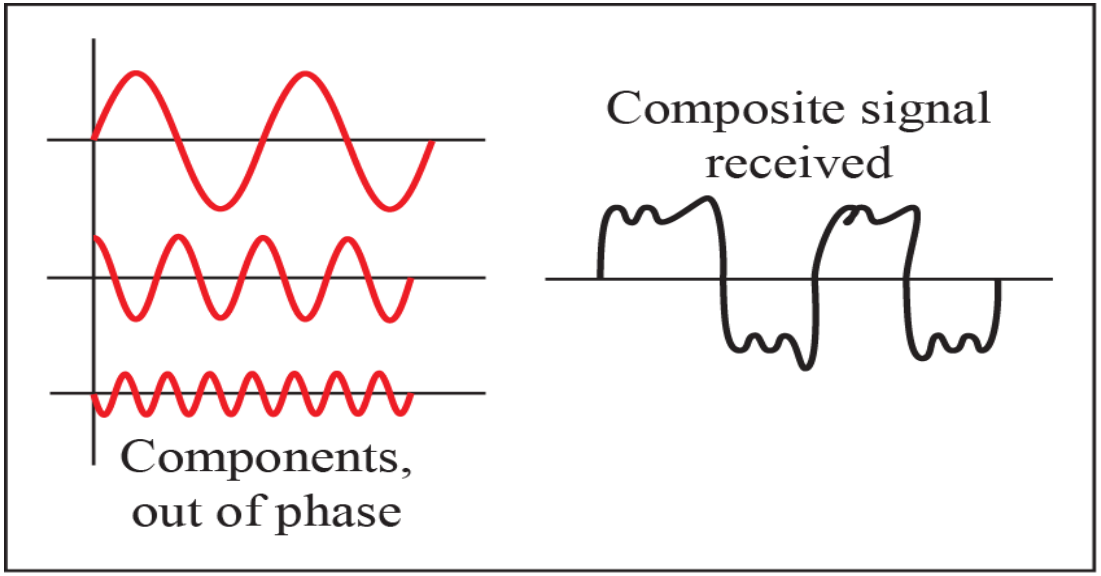
- *Each signal component has its own propagation speed through a medium and, therefore, its own delay in arriving at the final destination.*

- *Differences in delay may create a difference in phase if the delay is not exactly the same as the period duration.*

Distortion



At the sender

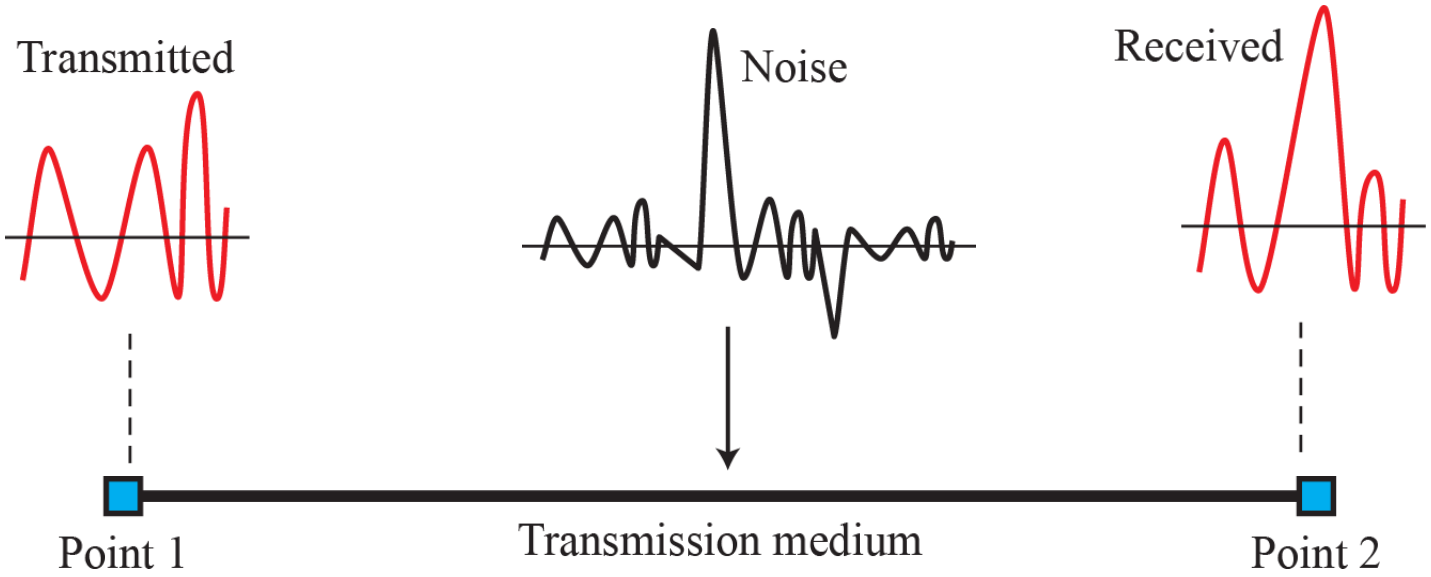


At the receiver

Noise

- *Noise is another cause of impairment.*
- *Several types of noise, such as thermal noise, induced noise, crosstalk, and impulse noise, may corrupt the signal.*
- *Thermal noise is the random motion of electrons in a wire, which creates an extra signal not originally sent by the transmitter.*
- *Induced noise comes from sources such as motors.*
- *Crosstalk is the effect of one wire on the other.*

Noise


Transmitted
Noise
Received
Point 1
Transmission medium
Point 2

Data Rate Limits

- *How fast we can send data, in bits per second, over a channel.*

- *Two theoretical formulas were developed to calculate the data rate:*

  Nyquist for a noiseless channel

  Shannon for a noisy channel.

Noiseless Channel: Nyquist Rate

For a noiseless channel, the Nyquist bit rate formula defines the theoretical maximum bit rate.

$$\text{BitRate} = 2 \times \text{bandwidth} \times \log_2 L$$

Noisy Channel: Shannon Capacity

In reality, we cannot have a noiseless channel; the channel is always noisy. In 1944, Claude Shannon introduced a formula, called the Shannon capacity, to determine the theoretical highest data rate for a noisy channel:

$$\text{Capacity} = \text{bandwidth} \times \log_2(1 + \text{SNR})$$

PERFORMANCE

One important issue in networking is the performance of the network— how good is it?

Bandwidth

- Bandwidth is a measure of how much data over time a communication link can handle, its capacity. This is typically measured as kbps (kilobits – thousands of bits per second), Mbps (megabits – millions of bits per second) or Gbps (gigabits – billions of bits per second)

- *can be used in two different contexts with two different measuring values:*

  bandwidth in hertz bandwidth in bits per second

Throughput

A measure of how fast we can actually send data through a network.

Throughput is the actual amount of data that is successfully sent/received over the communication link.

Throughput is presented as kbps, Mbps or Gbps, and can differ from bandwidth due to a range of technical issues, including latency, packet loss, jitter and more

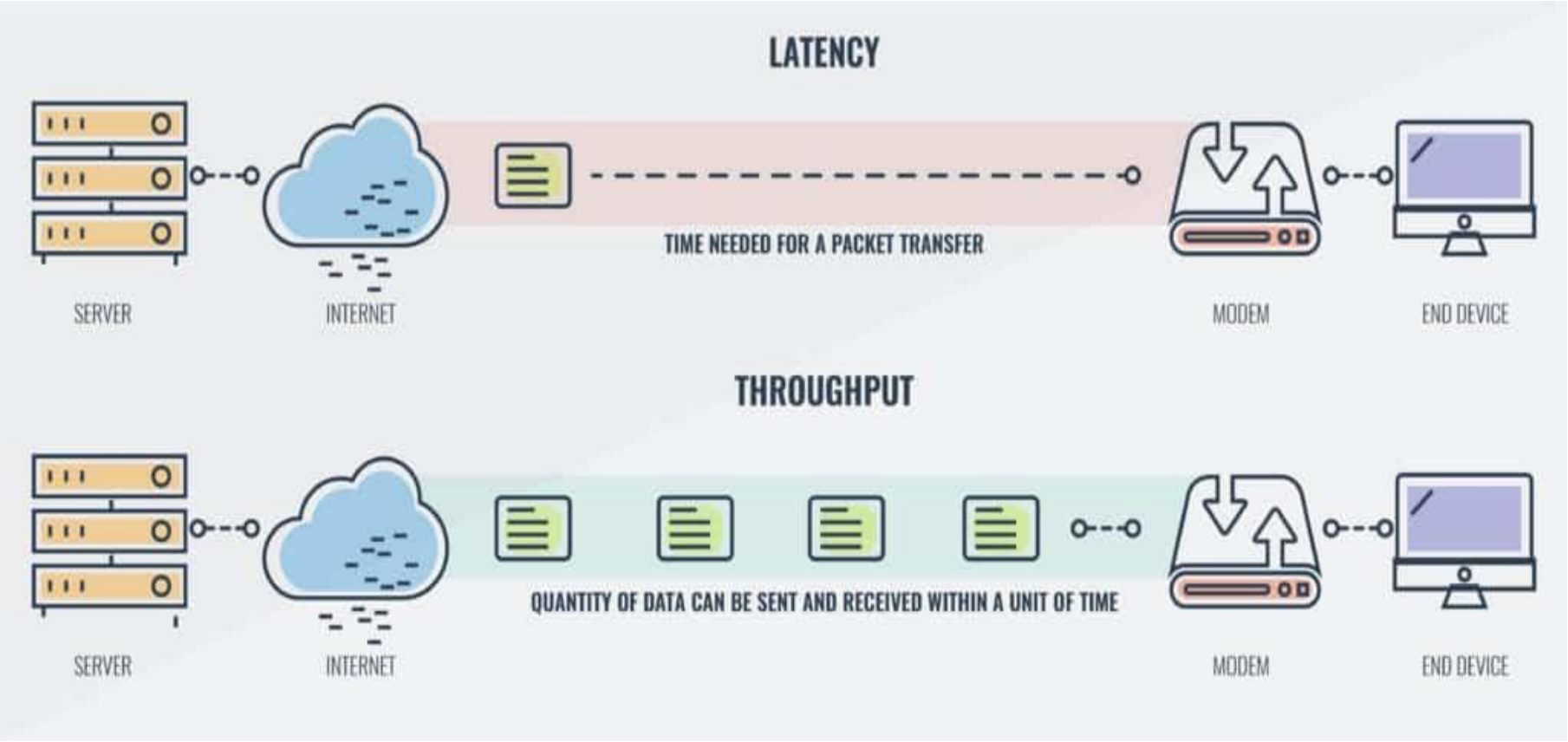
Latency

- *The latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source.*
- Latency is the time it takes for a packet to get across the network, from source to destination. It is measured in units of time — ms

- *We can say that latency is made of four components: propagation time, transmission time, queuing time and processing delay.*

Latency = propagation time + transmission time + queuing time + processing delay

Network Performance Measures



Example

A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

Solution

We can calculate the throughput as

$$\text{Throughput} = (12{,}000 \times 10{,}000) / 60 = 2 \text{ Mbps}$$

The throughput is almost one-fifth of the bandwidth in this case.

Jitter

- *Another performance issue that is related to delay is jitter.*

We can roughly say that jitter is a problem if different packets of data encounter different delays and the application using the data at the receiver site is time-sensitive (audio and video data, for example).

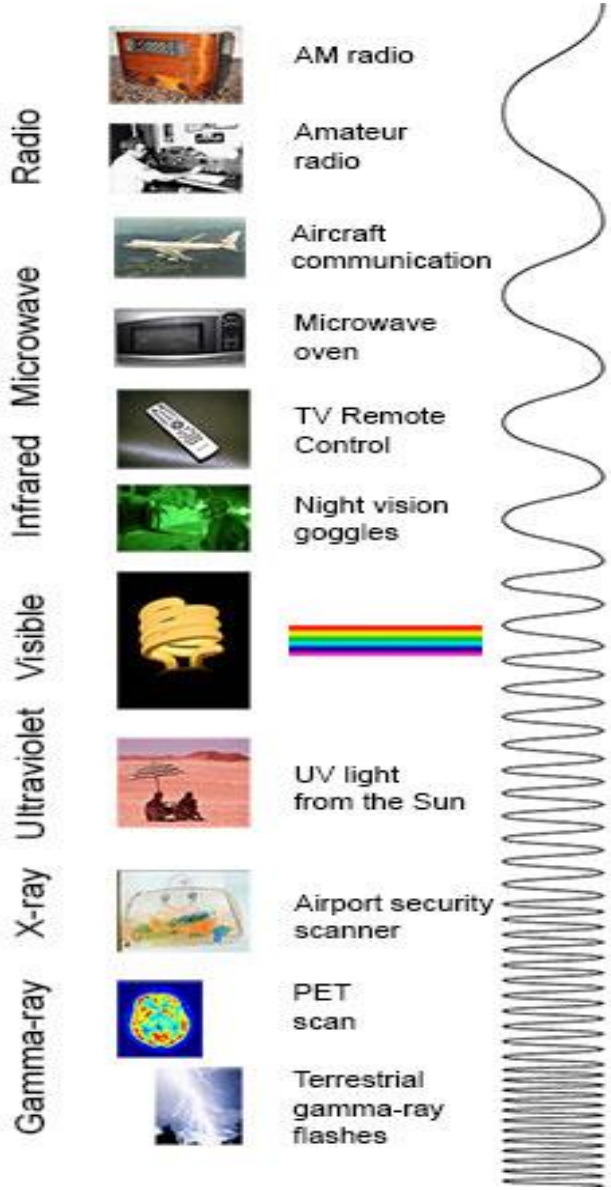
If the delay for the first packet is 20 ms, for the second is 45 ms, and for the third is 40 ms, then the real-time application that uses the packets endures jitter

Bandwidth Vs Speed

Speed refers to the rate at which data can be sent, while the definition of bandwidth is the capacity for that speed.
bandwidth capacity, and network speed

Bandwidth refers to the *maximum* capacity of an internet connection, not the actual speed.
For example, if you order a 100 Mbps package from your Internet Service Provider (ISP), your bandwidth would be 100 Mbps. That means the most data your connection can download at one time is 100 Mbps.
Your actual internet speed, on the other hand, is likely to be less than your bandwidth most of the time because of network congestion and other external factors.

Electromagnetic Spectrum



The electromagnetic (EM) spectrum is the range of all types of EM radiation

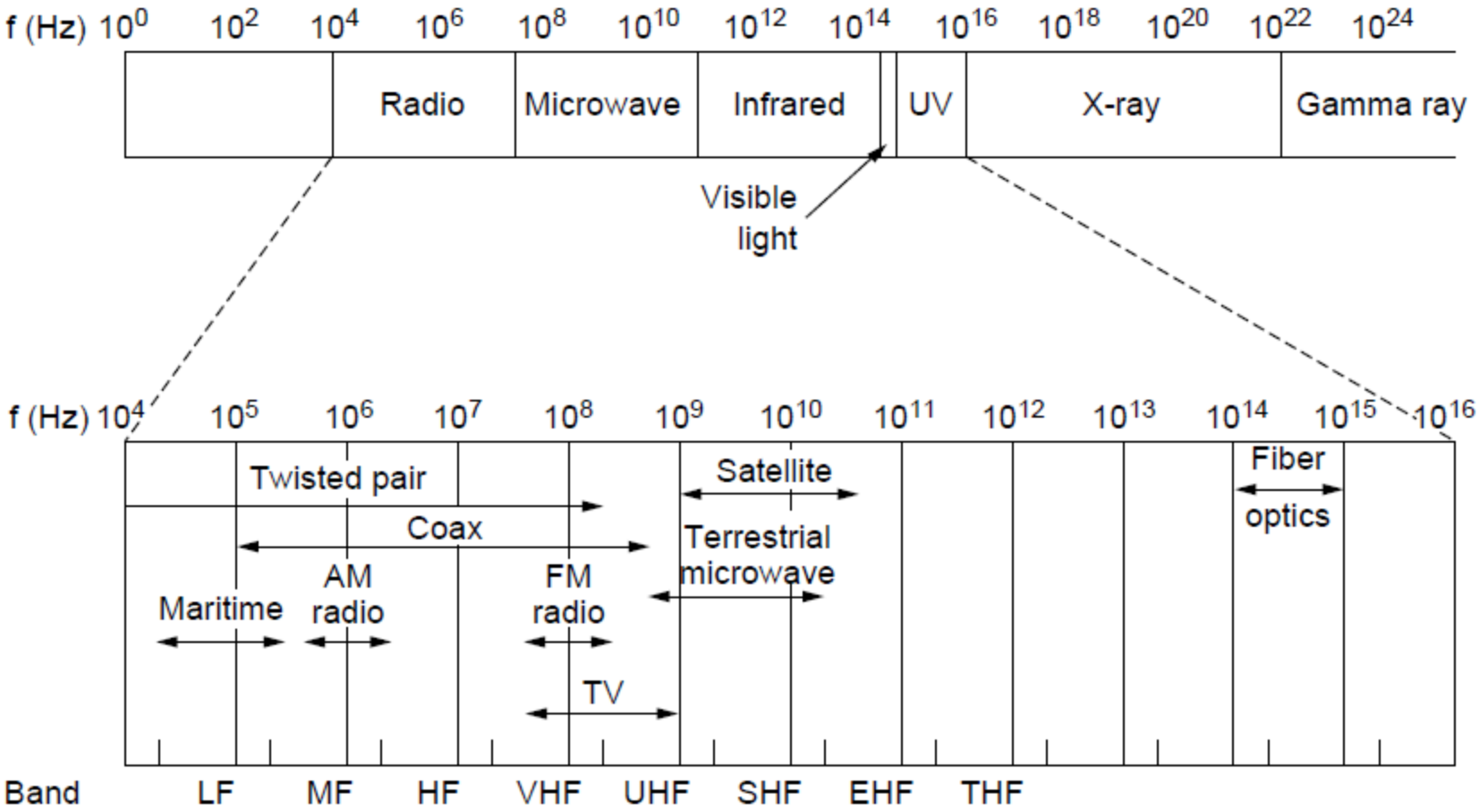
Radiation is energy that travels and spreads out as it goes – the visible light that comes from a lamp in your house and the radio waves that come from a radio station are two types of electromagnetic radiation.

The other types of EM radiation that make up the electromagnetic spectrum are microwaves, infrared light, ultraviolet light, X-rays and gamma-rays.

Electromagnetic Spectrum

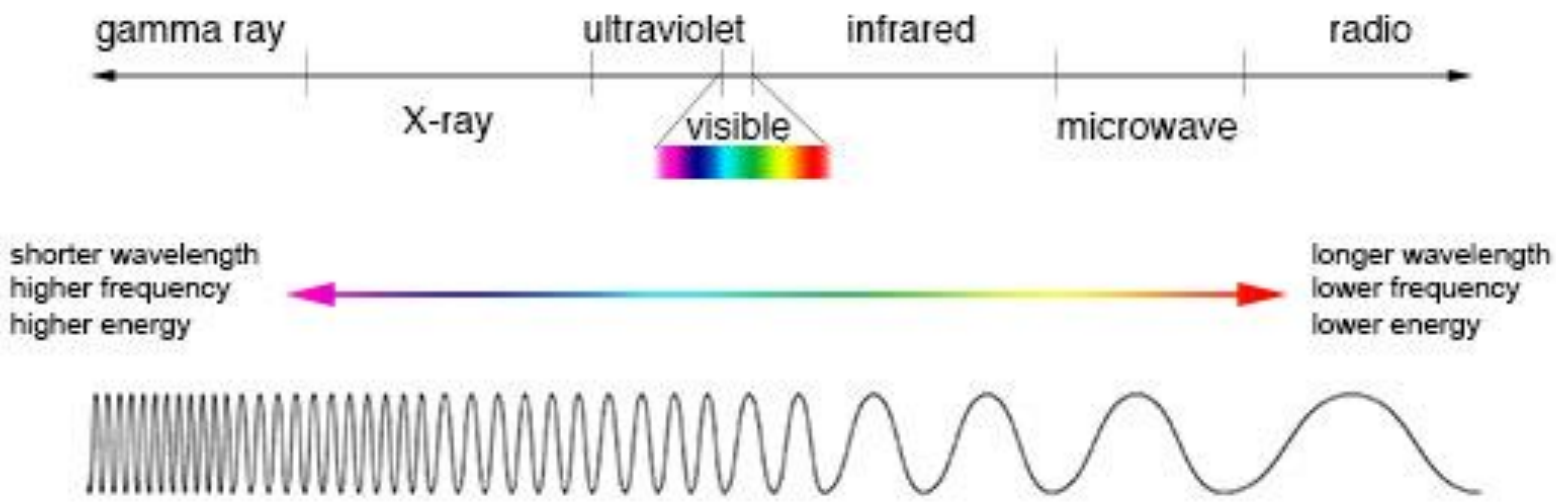
- **Radio:** Your radio captures radio waves emitted by radio stations, bringing your favorite tunes. Radio waves are also emitted by stars and gases in space.
- **Microwave:** Microwave radiation will cook your popcorn in just a few minutes, but is also used by astronomers to learn about the structure of nearby galaxies.
- **Infrared:** Night vision goggles pick up the infrared light emitted by our skin and objects with heat. In space, infrared light helps us map the dust between stars.
- **Visible:** Our eyes detect visible light. Fireflies, light bulbs, and stars all emit visible light.
- **Ultraviolet:** Ultraviolet radiation is emitted by the Sun and are the reason skin tans and burns. "Hot" objects in space emit UV radiation as well.
- **X-ray:** A dentist uses X-rays to image your teeth, and airport security uses them to see through your bag. Hot gases in the Universe also emit X-rays.
- **Gamma ray:** Doctors use gamma-ray imaging to see inside your body. The biggest gamma-ray generator of all is the Universe.

The Electromagnetic Spectrum (1)

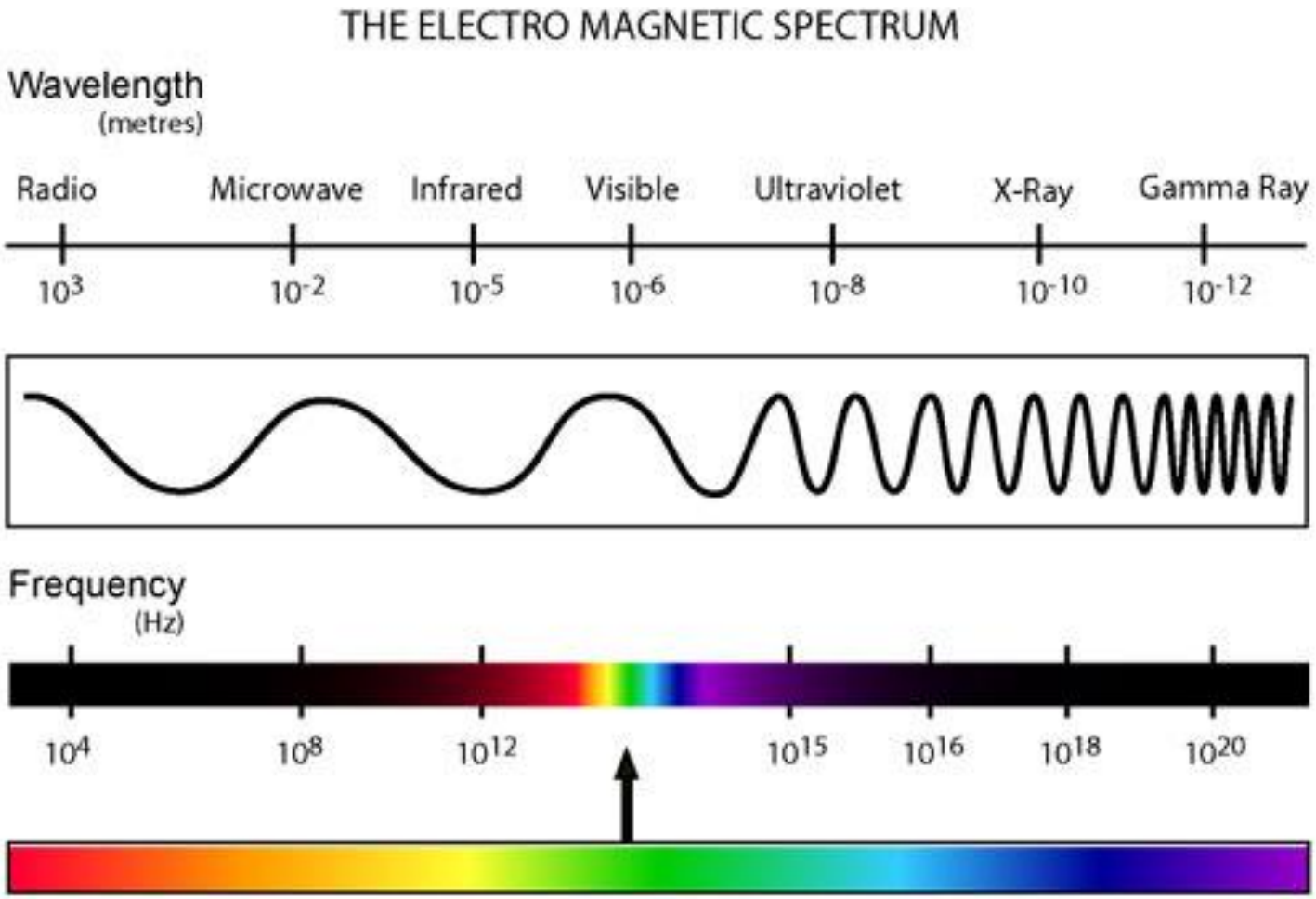


The electromagnetic spectrum and its uses for communication

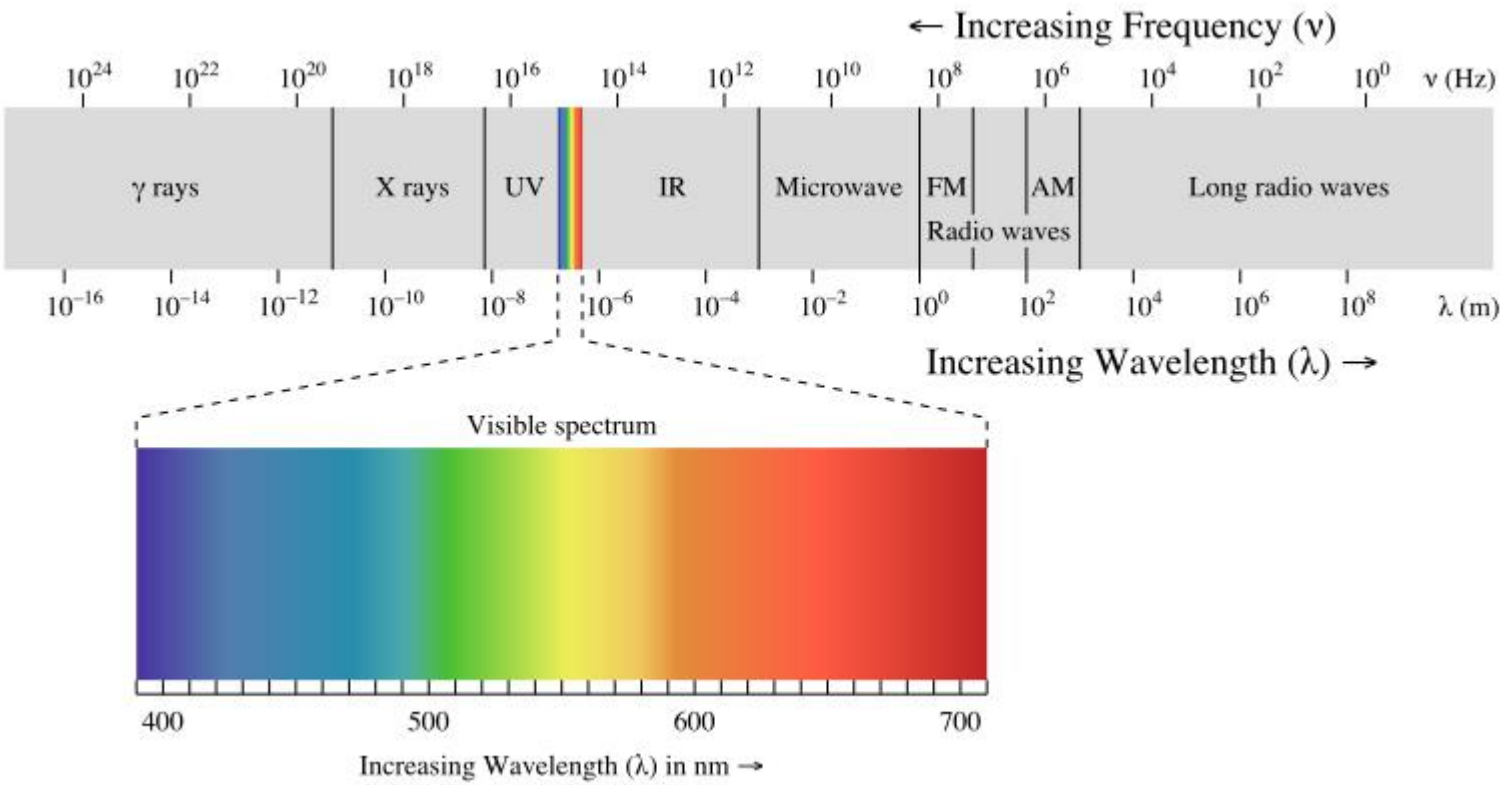
Electromagnetic Spectrum



Electromagnetic Spectrum



Electromagnetic Spectrum



Transmission Media

- **Guided media** - such as copper wire and fiber optics, and
- **Unguided media** - such as terrestrial wireless, satellite, and lasers through the air.

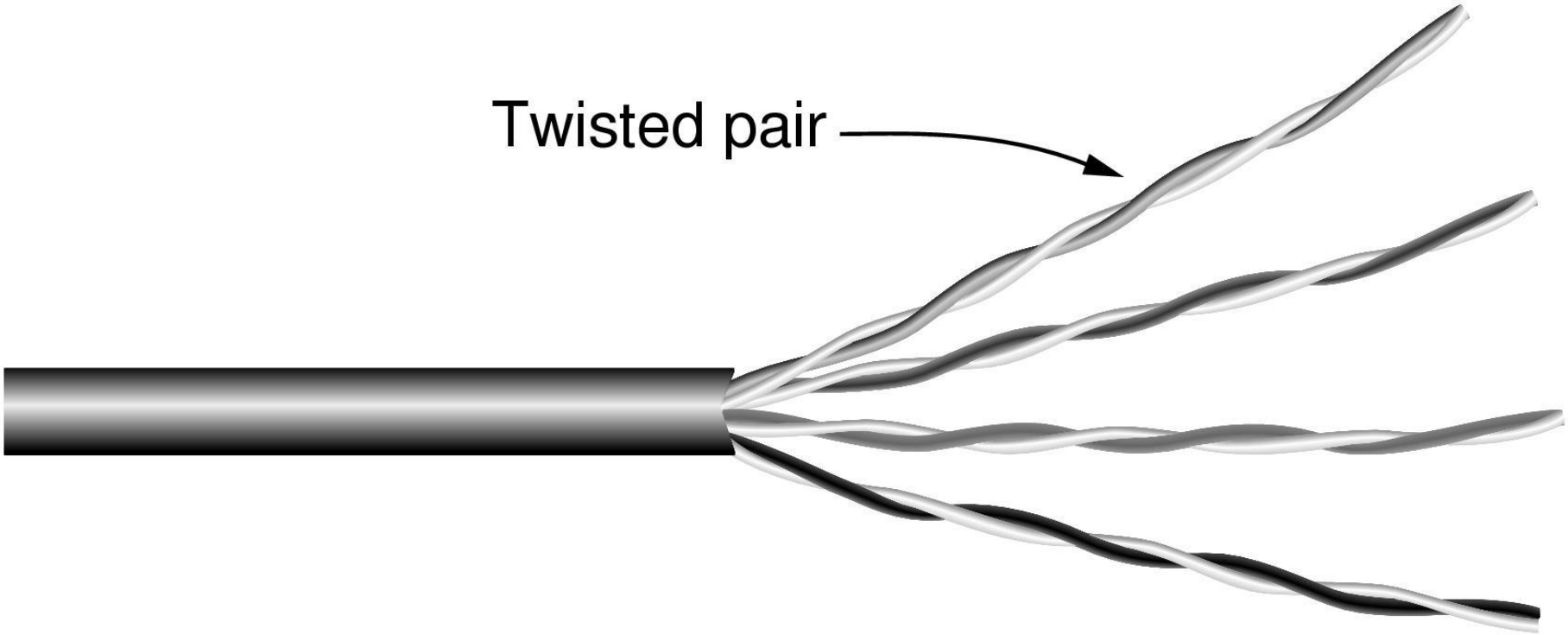
Guided Transmission Media

- Magnetic media
- Twisted pairs
- Coaxial cable
- Power lines
- Fiber optics

Guided Transmission Media

- Write data onto magnetic media
  - Magnetic tape or removable media (e.g., recordable DVDs)
- Although the bandwidth characteristics of magnetic tape are excellent, the delay characteristics are poor. Transmission time is measured in minutes or hours, not milliseconds

Twisted Pairs

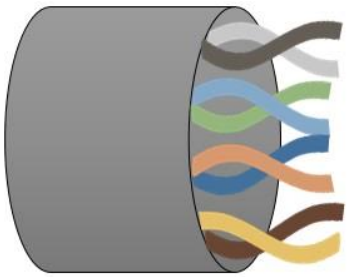


Category 5 UTP cable with four twisted pairs

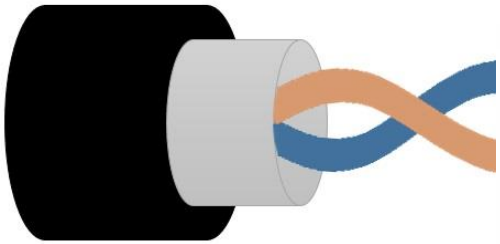
Twisted Pairs

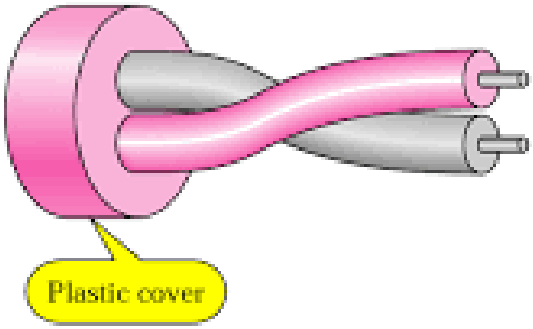
- One of the oldest and still most common transmission media is **twisted pair**

Unshielded Twisted Pair Cable

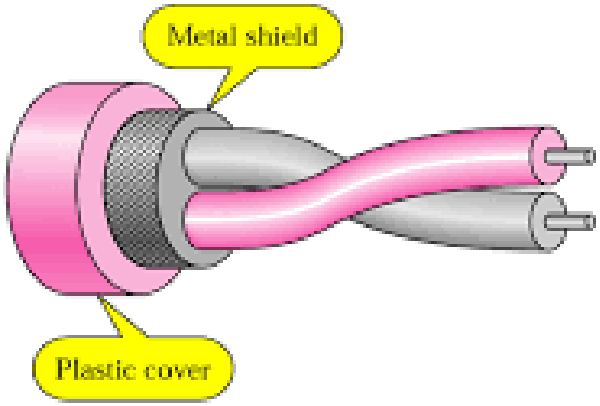


Shielded Twisted Pair Cable





a. UTP



b. STP

Twisted Pairs

- **Unshielded twisted-pair (UTP) cable** is the most prevalent type of telecommunication medium in use today. Its frequency range is suitable for transmitting both data and voice. Therefore, these are most commonly used in telephone systems.

  A twisted pair consists of two insulated conductors (usually copper) in a twisted configuration. Color bands are used in plastic insulation for identification.

- **Shielded twisted-pair (STP) cable** has an additional braided mesh coating or metal foil that wraps each set of insulated conductors. The metal casing intercepts the penetration of **electromagnetic noise**. It also can eradicate a phenomenon called crosstalk, which is the unwanted effect of one circuit (or channel) on another circuit (or channel).

Twisted Pairs

- Physical description:
  - Each wire with copper conductor
  - Separately insulated wires
  - Twisted together to reduce cross talk
  - Often bundled into cables of two or four twisted pairs
  - If enclosed in a sheath then is shielded twisted pair (STP) otherwise often for home usage unshielded twisted pair (UTP). Must be shield from voltage lines
- Application:
  - Common in building for digital signaling used at speed of 10's Mb/s (CAT3) and 100Mb/s (CAT5) over 100s meters.
  - Common for telephone interconnection at home and office buildings
  - Less expensive medium; limited in distance, bandwidth, and data rate

Twisted Pairs

UTP Categories - Copper Cable				
UTP Category	**Data Rate**	**Max. Length**	**Cable Type**	**Application**
CAT1	Up to 1Mbps	-	Twisted Pair	Old Telephone Cable
CAT2	Up to 4Mbps	-	Twisted Pair	Token Ring Networks
CAT3	Up to 10Mbps	100m	Twisted Pair	Token Rink & 10BASE-T Ethernet
CAT4	Up to 16Mbps	100m	Twisted Pair	Token Ring Networks
CAT5	Up to 100Mbps	100m	Twisted Pair	Ethernet, FastEthernet, Token Ring
CAT5e	Up to 1 Gbps	100m	Twisted Pair	Ethernet, FastEthernet, Gigabit Ethernet
CAT6	Up to 10Gbps	100m	Twisted Pair	GigabitEthernet, 10G Ethernet (55 meters)
CAT6a	Up to 10Gbps	100m	Twisted Pair	GigabitEthernet, 10G Ethernet (55 meters)
CAT7	Up to 10Gbps	100m	Twisted Pair	GigabitEthernet, 10G Ethernet (100 meters)

Twisted Pairs

CAT5e

CAT6

Cat5e

Cat6

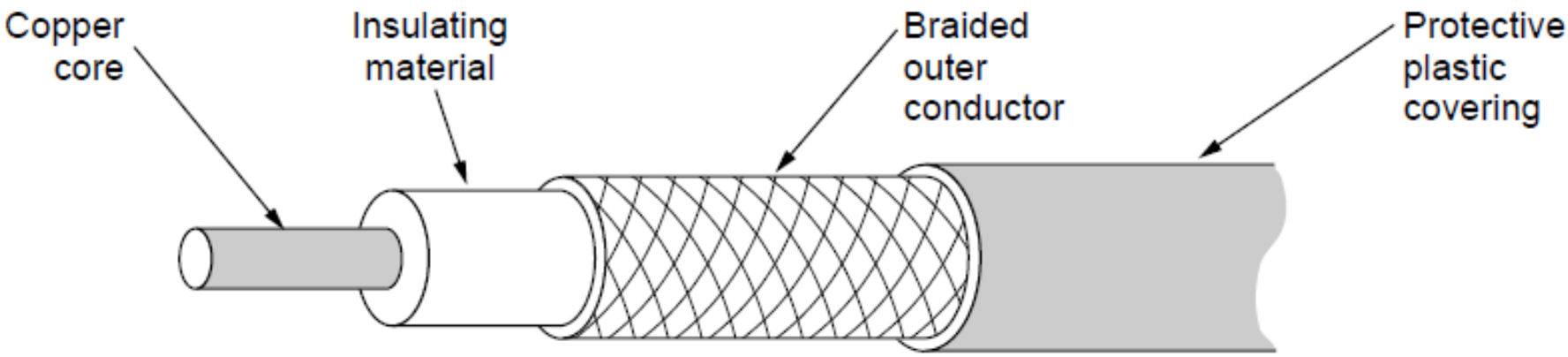
Cat6a

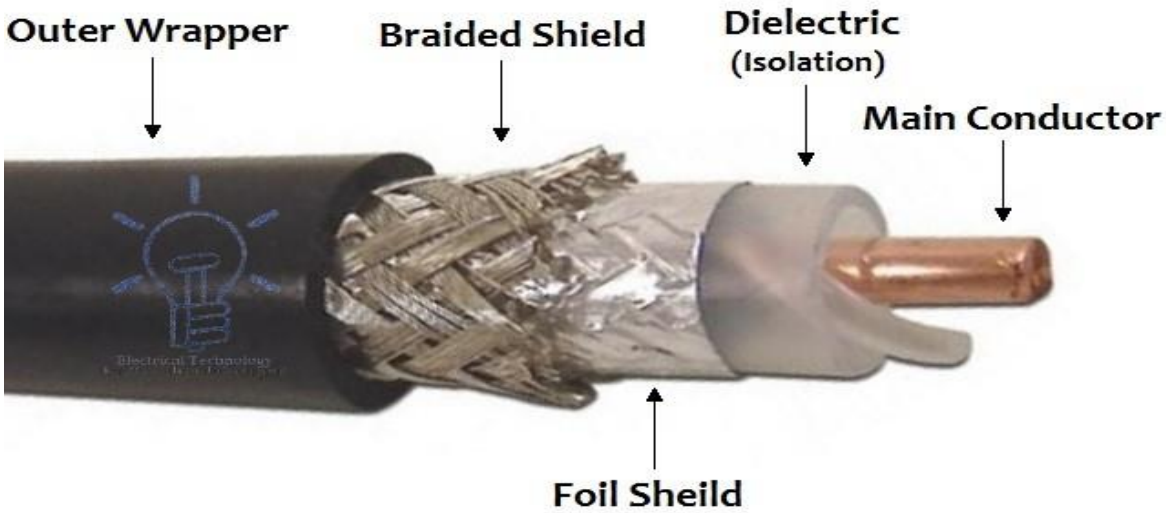
Cat7

Twisted Pairs

- **Cat 5** replaced earlier Category 3 cables with a similar cable that uses the same connector, but has more twists per meter. **More twists** result in less crosstalk and a better-quality signal over longer distances, making the cables more suitable for high-speed computer communication, especially 100-Mbps and 1-Gbps Ethernet LANs
- Through Category 6, these wiring types are referred to as **UTP (Unshielded Twisted Pair)** as they consist simply of wires and insulators. In contrast to these, Category 7 cables have shielding on the individual twisted pairs, as well as around the entire cable

Coaxial Cable





Coaxial Cable

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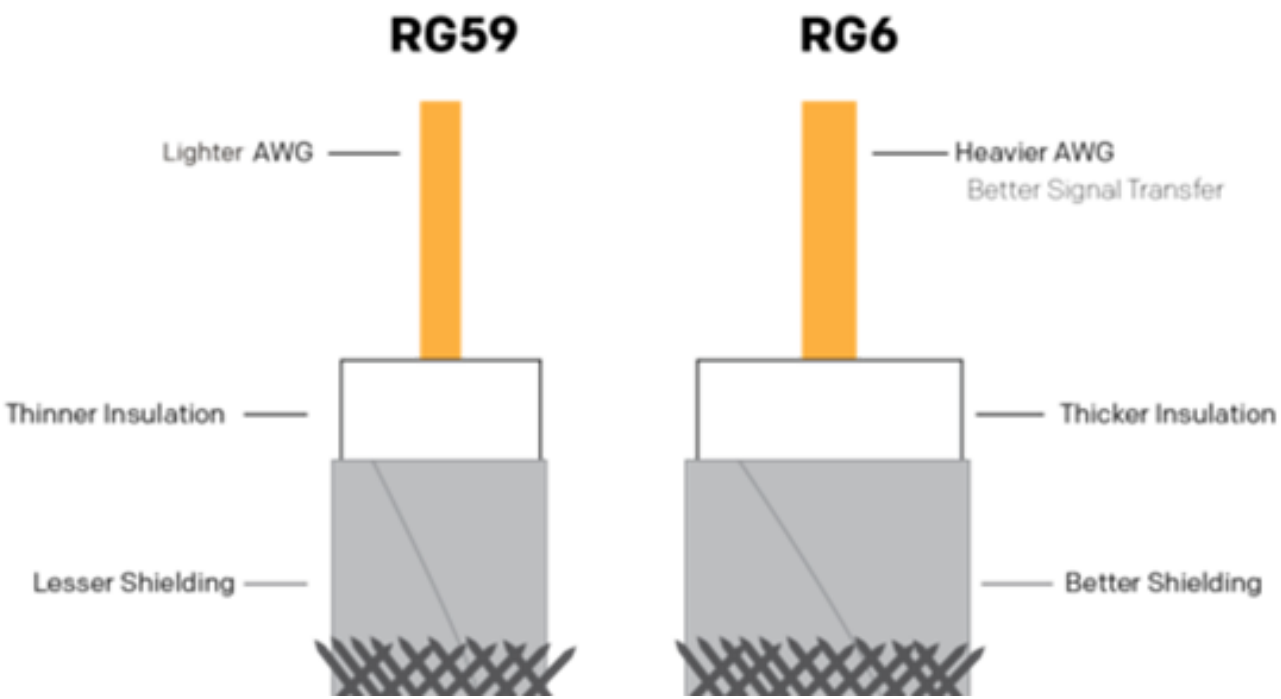
Coaxial Cable

- sometimes known as coax cable, is an electrical cable which transmits radio frequency (RF) signals from one point to another
- Coaxial cables are a popular choice because their shielded design allows the centre conductor to transmit data quickly while being protected from damage and interference. The transmission speed of coaxial cable is 10Mbps (megabits per second), and they offer 80 times more transmission capacity than twisted pair cables
- Coaxial cables are mainly built up of these four different layers:
- 1. A centre conductor which is usually a copper wire, which data and video travels through
- 2. Surrounding the copper wire is a dielectric plastic insulator
- 3. A braided mesh made from copper then helps to shield the cable from electromagnetic interference (EMI)
- 4. The external layer is a plastic coating which protects the internal layers from damage
- Cable TVs, LAN

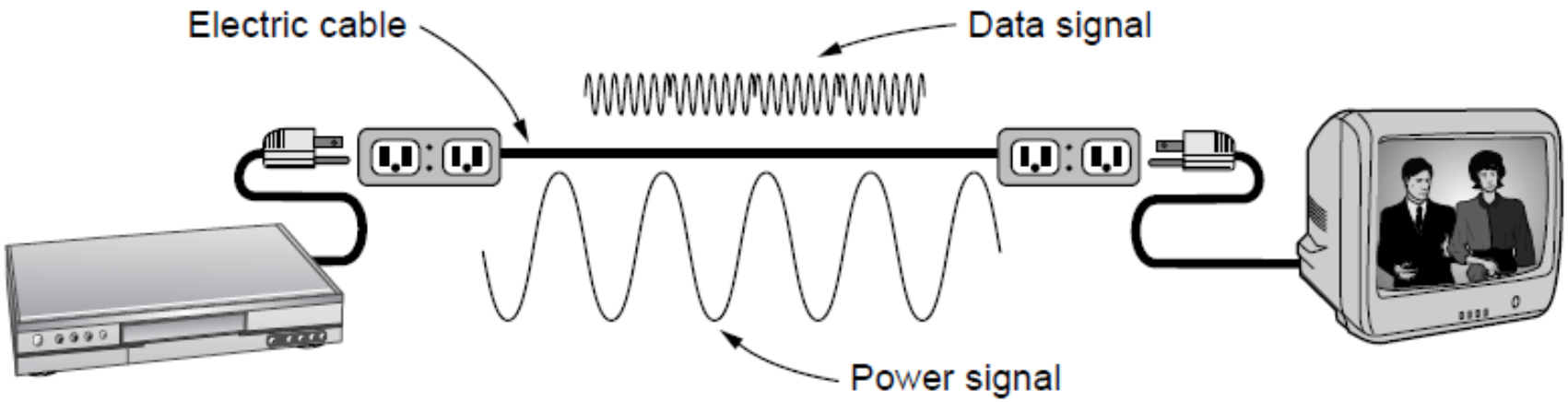
Coaxial Cable

Coaxial Cable

- RG6 cable is heavier gauge and has insulation and shielding tuned for high-bandwidth, high-frequency applications such as Internet, Cable TV, and Satellite TV signals.
- RG59 cable is thinner and is recommended in low bandwidth and lower frequency applications such as analog video and CCTV installations
- RG-Radio Guide, 6, 59 – randomly assigned number

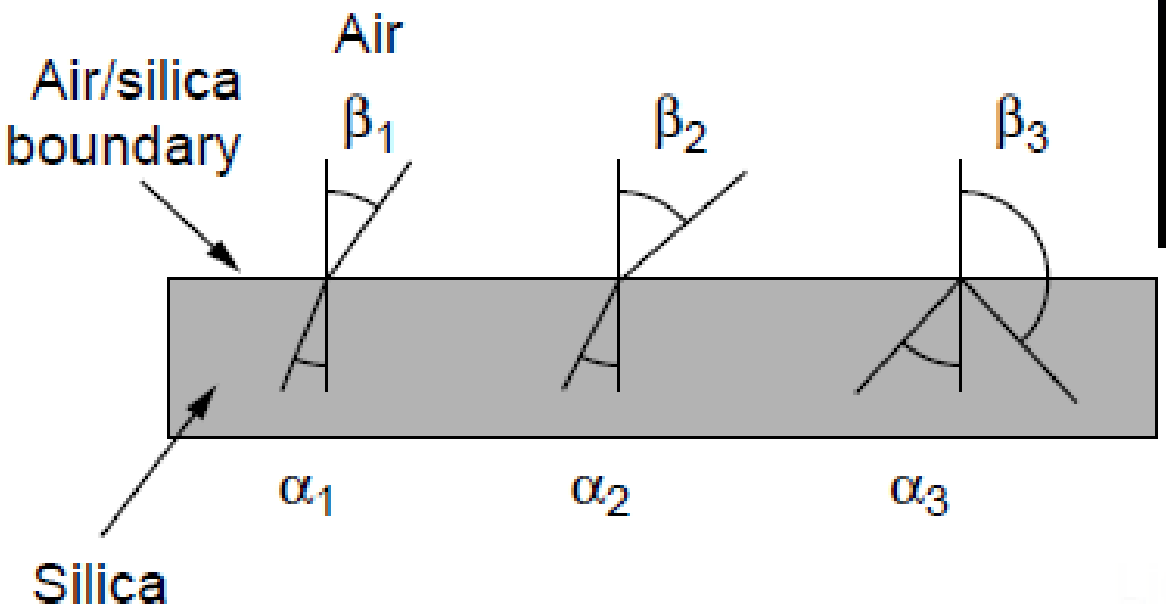


Power Lines



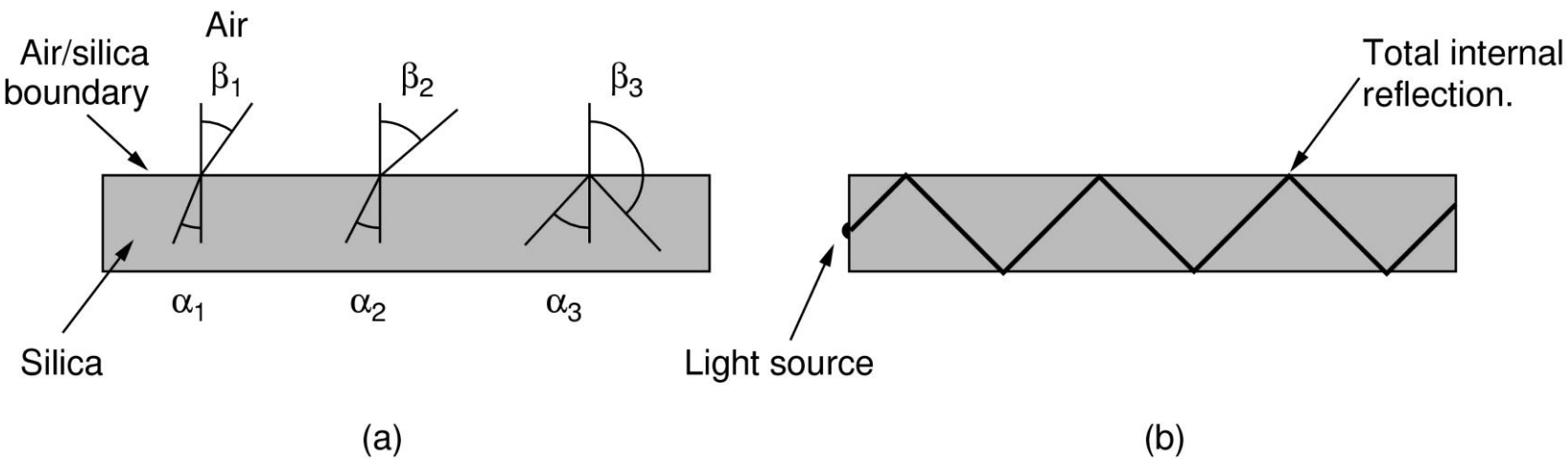
A network that uses household electrical wiring.

Fiber Optics (1)



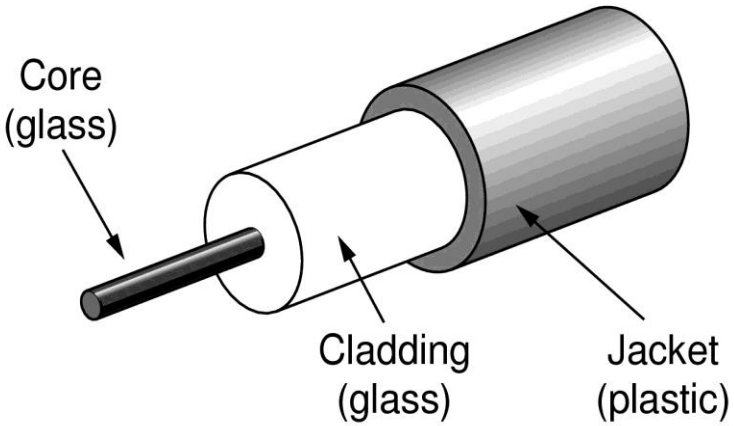
Three examples of a light ray from inside a silica fiber impinging on the air/silica boundary at different angles.

Fiber Optics (2)

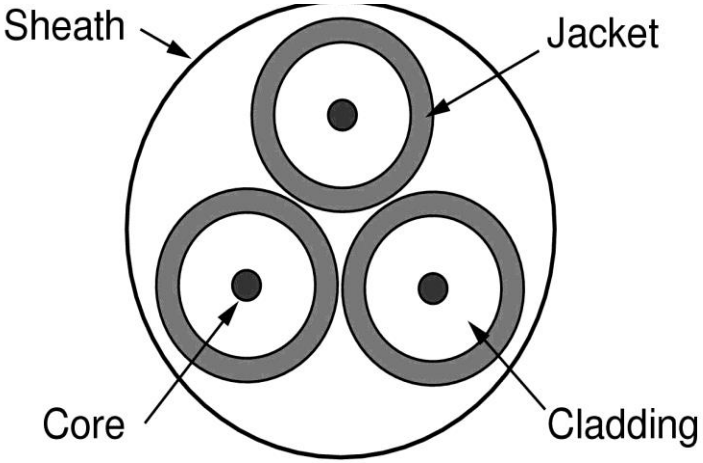


Light trapped by total internal reflection.

Fiber Optics (2)

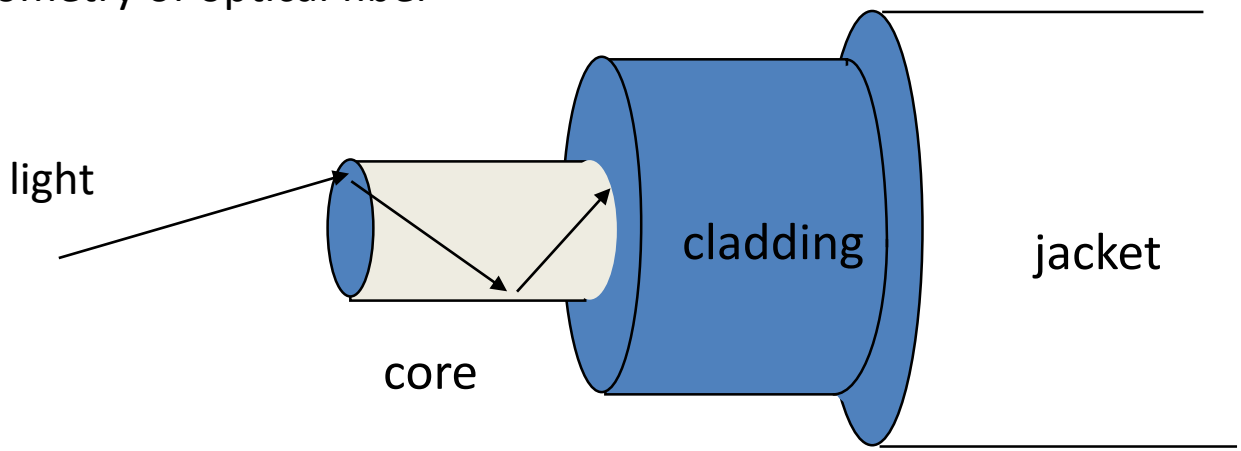


(a)

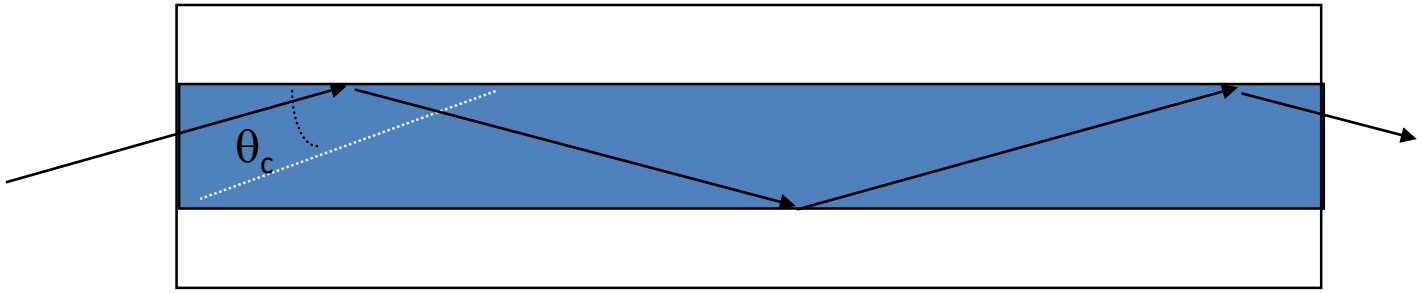


(b)

(a) Geometry of optical fiber



(b) Reflection in optical fiber

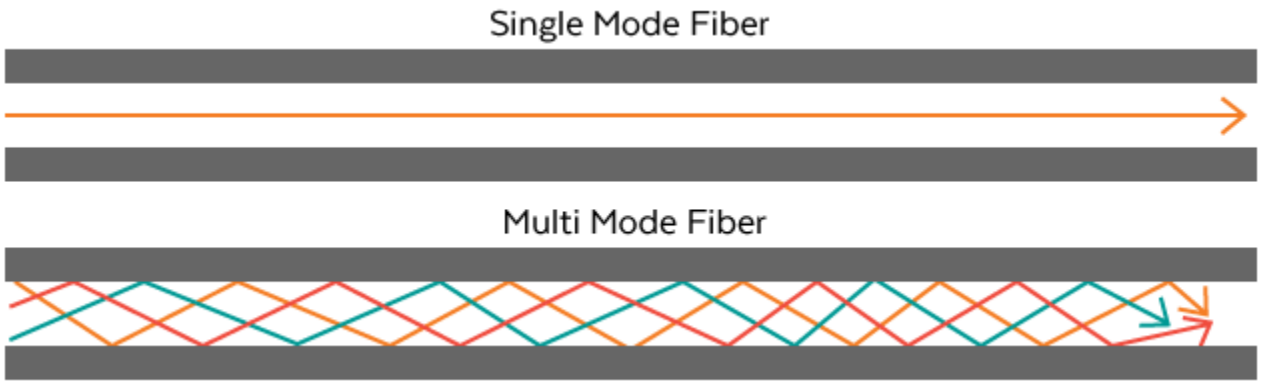


Fiber Optics (2)

- Physical Description:
  - Glass or plastic core of optical fiber = 2to125 µm
  - Cladding is an insulating material
  - Jacket is a protective cover
  - Laser or light emitting diode provides transmission light source
- Applications:
  - Long distance telecommunication
  - Greater capacity; 2 Gb/s over 10's of Km
  - Smaller size and lighter weight
  - Lower attenuation (reduction in strength of signal)
  - Electromagnetic isolation – not effected by external electromagnetic environment. Aka more privacy
  - Greater repeater spacing – fewer repeaters, reduces line regeneration cost

Fiber Optics (2)

- Multimode fiber is optical fiber that is designed to carry multiple light rays or modes concurrently, each at a slightly different reflection angle within the optical fiber core. used for relatively short distances because the modes tend to disperse over longer lengths (this is called modal dispersion)
- For longer distances, single mode fiber (sometimes called monomode) fiber is used. In single mode fiber a single ray or mode of light act as a carrier



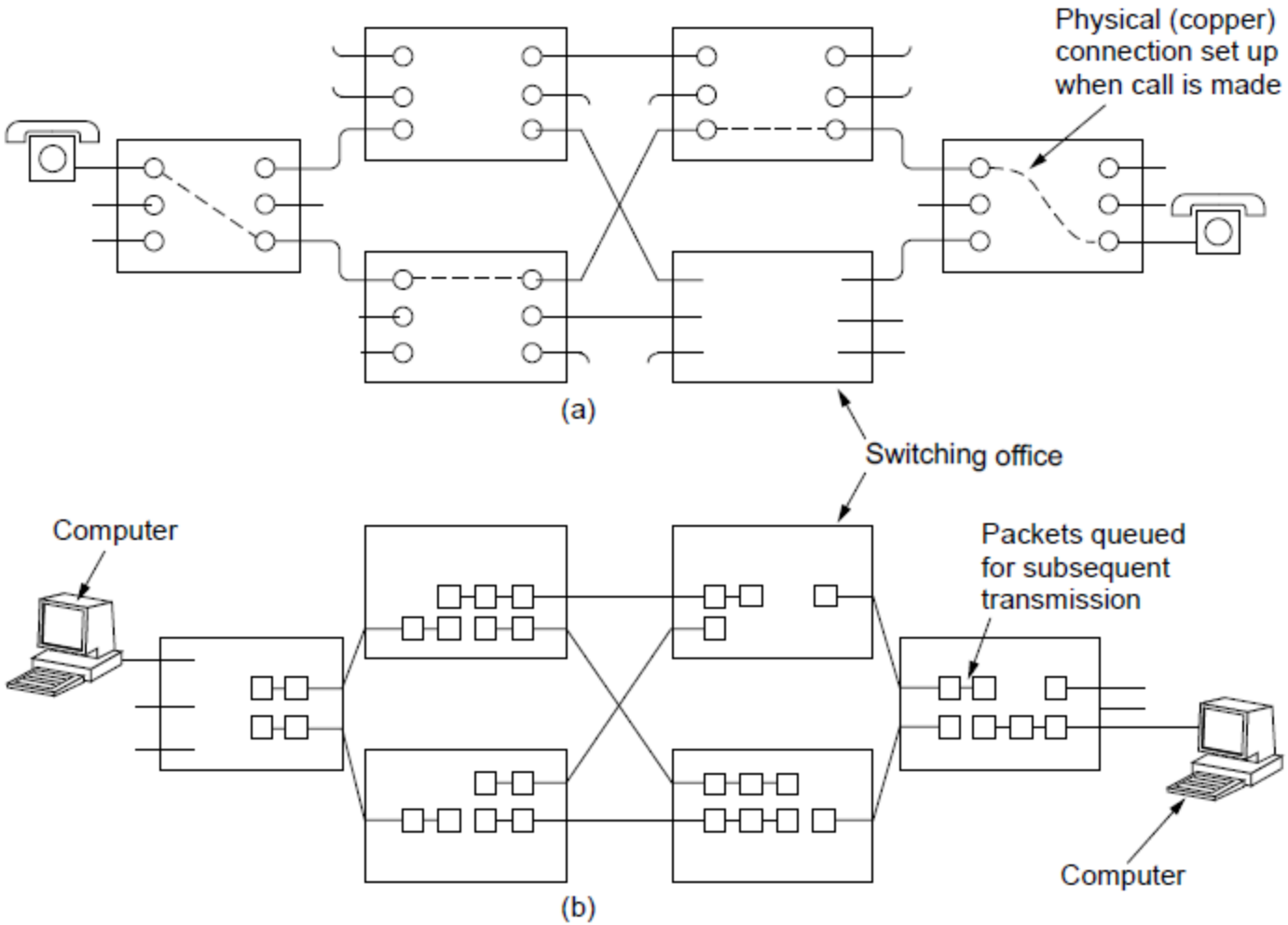
Fiber Optics (2)

- **Basic elements of Fiber optic cable:**
- **Core:** The optical fiber consists of a narrow strand of glass or plastic known as a core. A core is a light transmission area of the fiber. The more the area of the core, the more light will be transmitted into the fiber.
- **Cladding:** The concentric layer of glass is known as cladding. The main functionality of the cladding is to provide the lower refractive index at the core interface as to cause the reflection within the core so that the light waves are transmitted through the fiber.
- **Jacket:** The protective coating consisting of plastic is known as a jacket. The main purpose of a jacket is to preserve the fiber strength, absorb shock and extra fibre protection.

Fiber Optics (2)

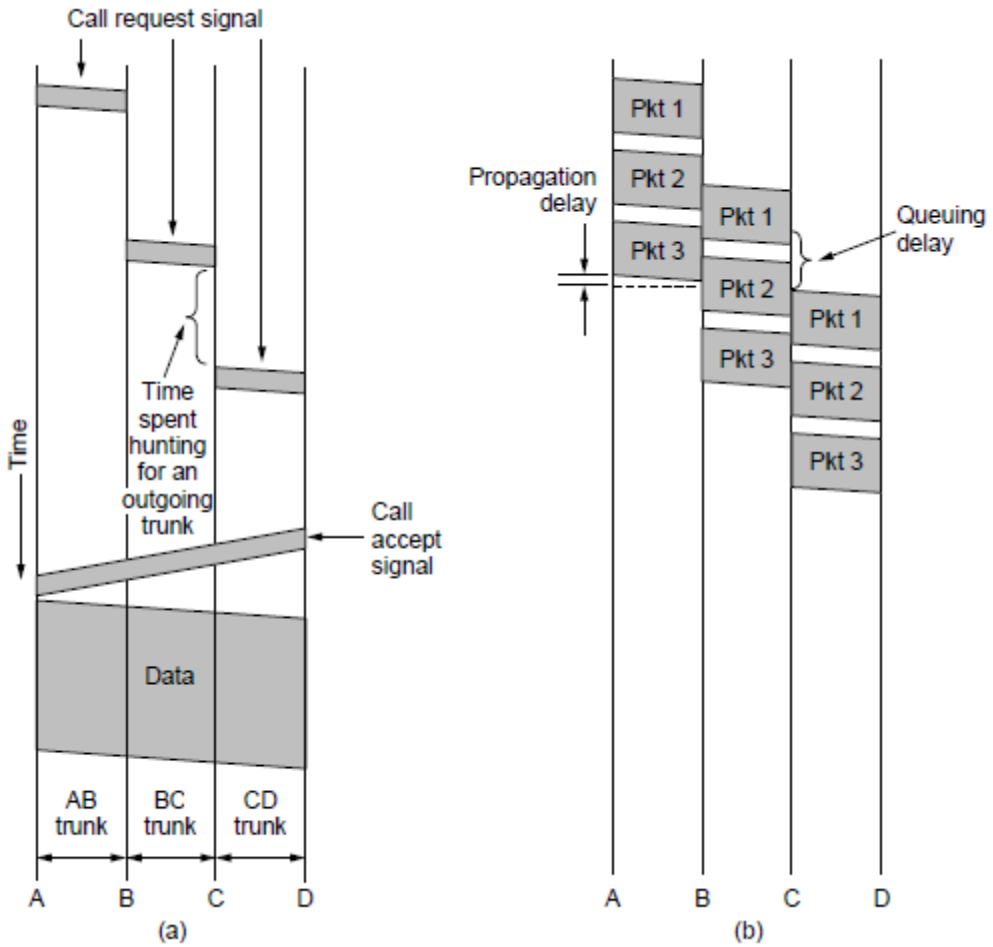
- **Following are the advantages of fiber optic cable over copper:**
- **Greater Bandwidth:** The fiber optic cable provides more bandwidth as compared copper. Therefore, the fiber optic carries more data as compared to copper cable.
- **Faster speed:** Fiber optic cable carries the data in the form of light. This allows the fiber optic cable to carry the signals at a higher speed.
- **Longer distances:** The fiber optic cable carries the data at a longer distance as compared to copper cable.
- **Better reliability:** The fiber optic cable is more reliable than the copper cable as it is immune to any temperature changes while it can cause obstruct in the connectivity of copper cable.
- **Thinner and Sturdier:** Fiber optic cable is thinner and lighter in weight so it can withstand more pull pressure than copper cable.

Circuit Switching/Packet Switching (1)



(a) Circuit switching. (b) Packet switching.

Circuit Switching/Packet Switching (2)



Timing of events in (a) circuit switching, (b) packet switching

Circuit Switching/Packet Switching (3)

Item	Circuit switched	Packet switched
Call setup	Required	Not needed
Dedicated physical path	Yes	No
Each packet follows the same route	Yes	No
Packets arrive in order	Yes	No
Is a switch crash fatal	Yes	No
Bandwidth available	Fixed	Dynamic
Time of possible congestion	At setup time	On every packet
Potentially wasted bandwidth	Yes	No
Store-and-forward transmission	No	Yes
Charging	Per minute	Per packet

A comparison of circuit-switched and packet-switched networks.